

Programmable Logic Controllers

Third Edition

Chapter 7

Programming Timers

Timers

There are very few industrial control systems that do not need at least one or two timed functions. They are used to activate or de-activate a device after a preset interval of time.

Time delay relays and solid-state timers are used to provide a time delay. They may have displays, pots or other means of operator interface for time settings and electromechanical or solid state outputs.

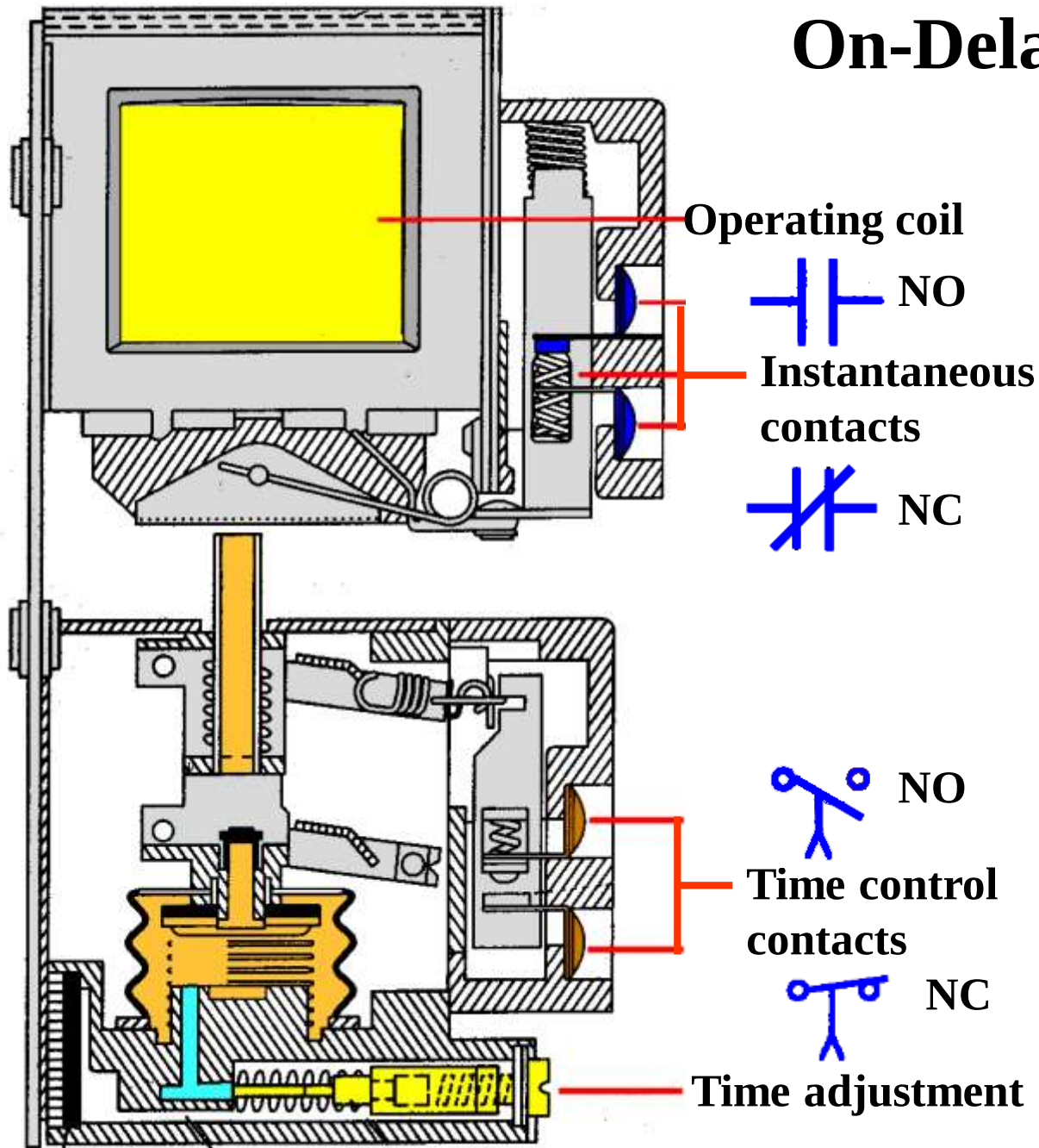
**Time Delay
Relay**



**Solid-State
Timer**



On-Delay Timing Relay



Nontimed contacts are controlled directly by the timer coil, as in a general-purpose control relay.

When the coil is energized, the timed contacts are prevented from opening or closing until the time delay period has elapsed. However, when the coil is de-energized, the timed contacts return instantaneously to their normal state.

Timed Contact Symbols

On-Delay Symbols



**Normally open, timed
closed contact (NOTC)**

**Contact is open when
relay coil is de-energized**

**When relay is energized,
there is a time delay in
closing**



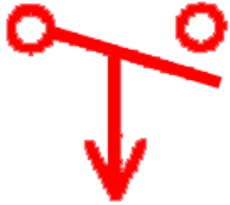
**Normally closed, timed
open contact (NCTO)**

**Contact is closed when
relay coil is de-energized**

**When relay is energized,
there is a time delay in
opening**

Timed Contact Symbols

Off Delay Symbols



Normally open, timed open contacts (NOTO).

Contact is normally open when relay coil is de-energized.

When relay coil is energized, contact closes instantly.

When relay coil is de-energized, there is a time delay before the contact opens.



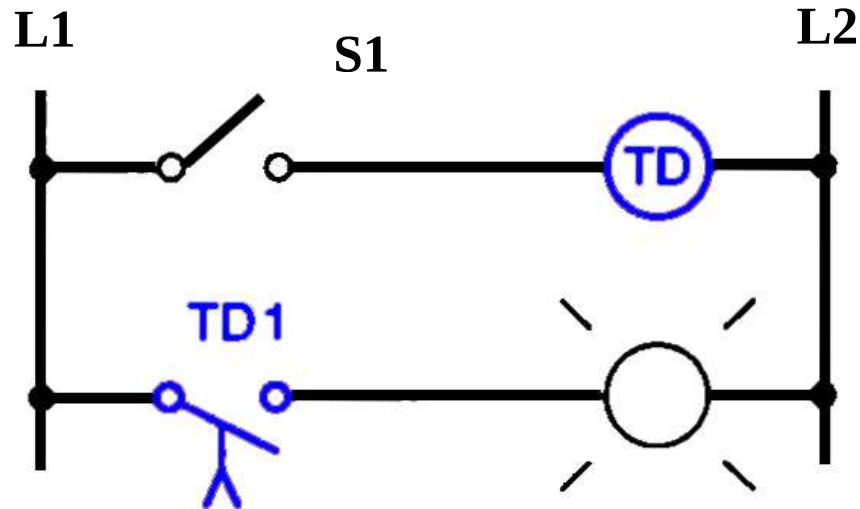
Normally closed, timed closed contacts (NCTC).

Contact is normally closed when relay coil is de-energized.

When relay coil is energized, contact opens instantly.

When relay coil is de-energized, there is a time delay before the contact closes.

On-Delay Relay Timer Circuit (NOTC Contact)

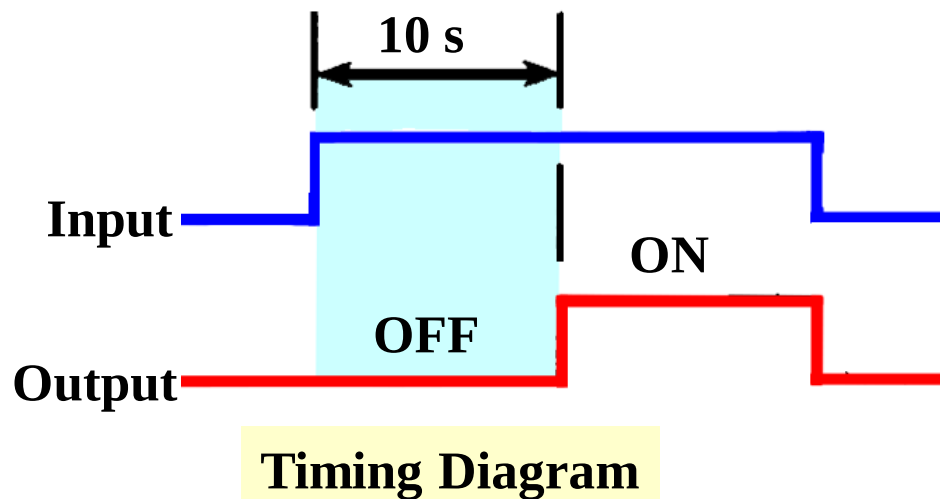


Sequence of operation

S1 open, TD de-energized, TD1 open, L1 is off.

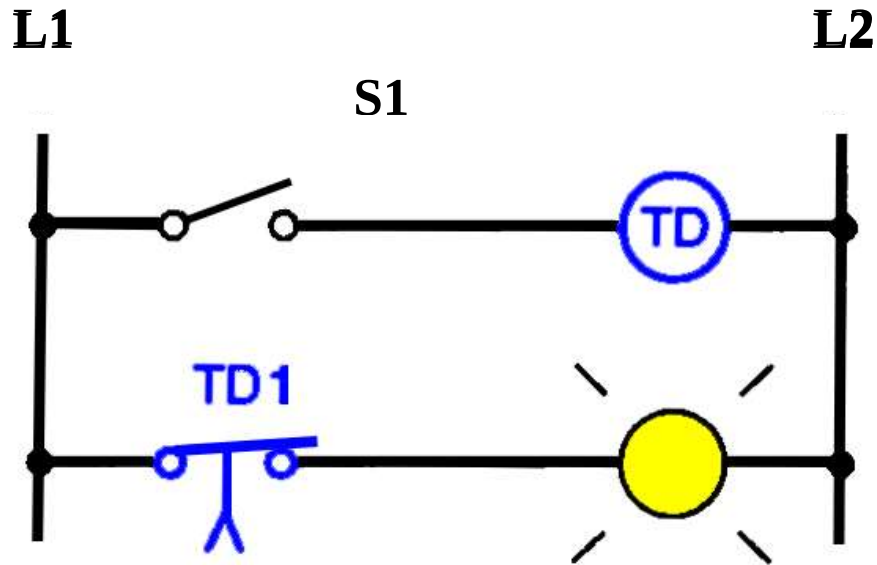
S1 closes, TD energizes, timing period starts, TD1 still open, L1 is still off.

After 10 s, TD1 closes, L1 is switched on.



S1 is opened, TD de-energizes, TD1 opens instantly, L1 is switched off.

On-Delay Relay Timer Circuit (NCTO Contact)

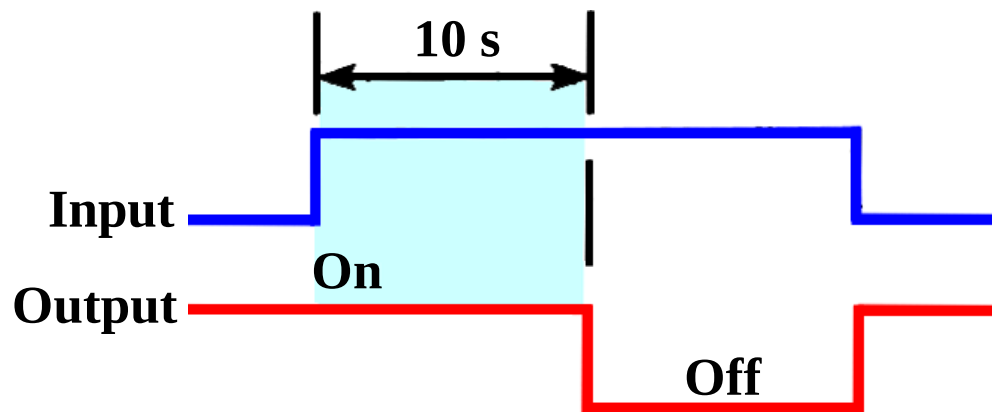


Sequence of operation

S1 open, TD de-energized, TD1 closed, L1 is on.

S1 closes, TD energizes, timing period starts, TD1 is still closed, L1 is still on.

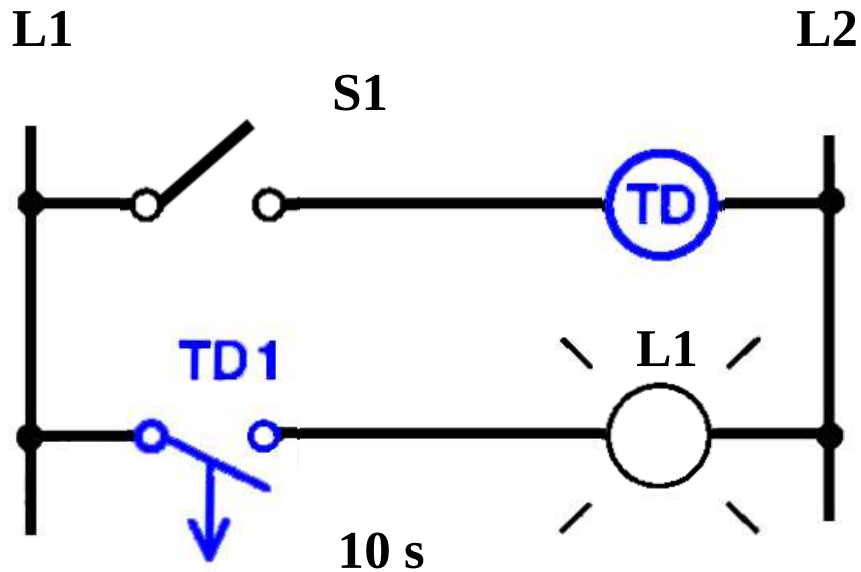
After 10 s, TD1 opens, L1 is switched off.



Timing Diagram

S1 is opened, TD de-energizes, TD1 closes instantly, L1 is switched on.

Off-Delay Relay Timer Circuit (NOTO Contact)



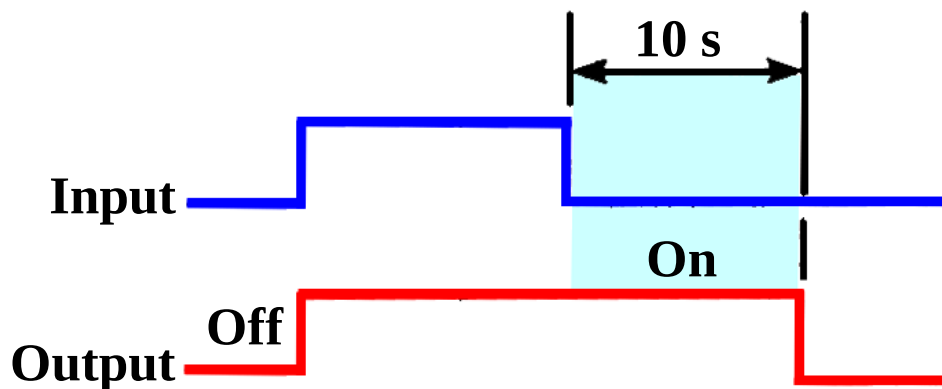
Sequence of operation

S1 open, TD de-energized, TD1 open, L1 is off.

S1 closes, TD energizes, TD1 closes instantly, L1 is switched on.

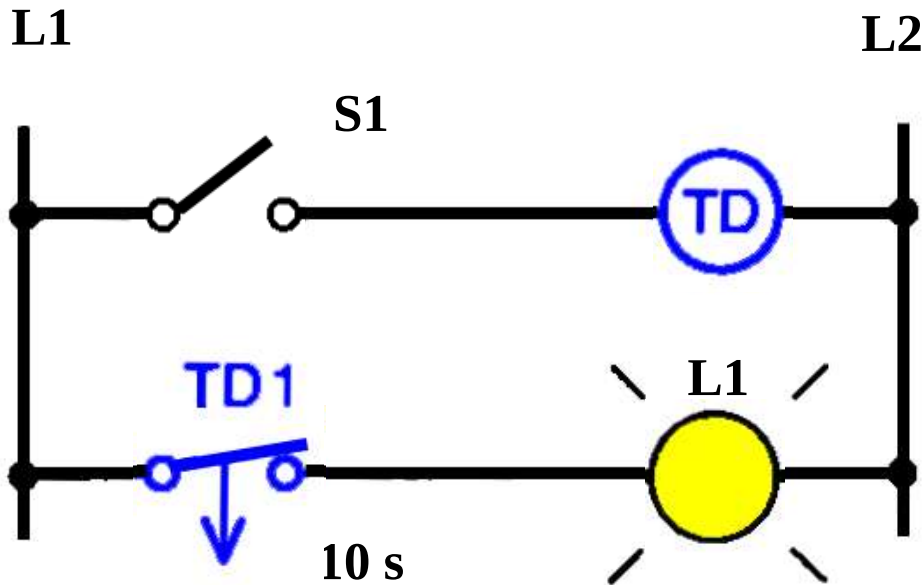
S1 is opened, TD de-energizes, timing period starts, TD1 is still closed, L1 is still on.

After 10 s, TD1 opens, L1 is switched off.



Timing Diagram

Off-Delay Relay Timer Circuit (NCTC Contact)



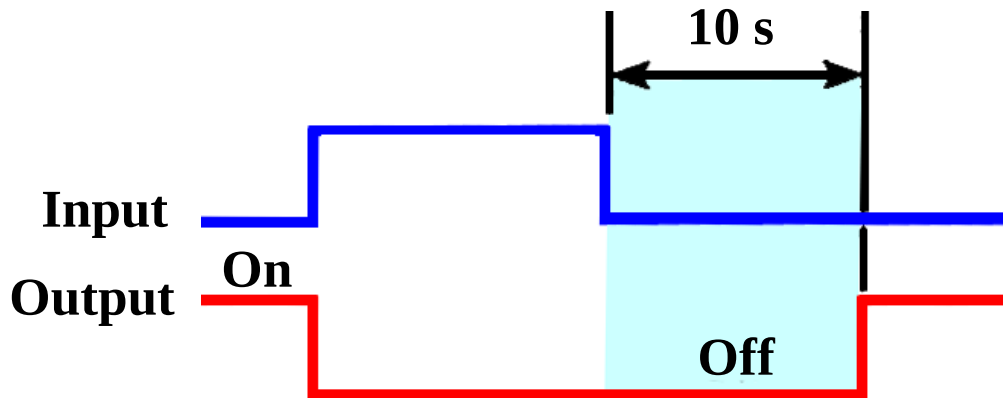
Sequence of operation

S1 open, TD de-energized, TD1 closed, L1 is on.

S1 closes, TD energizes, TD1 opens instantly, L1 is switched off.

S1 is opened, TD de-energizes, timing period starts, TD1 is still open, L1 is still off.

After 10 s, TD1 closes, L1 is switched on.



Timing Diagram

1. Timers are used to activate or de-activate a device after a preset interval of time.

(True/False)

2. With most timers the time delay period is fixed and can not be varied. (True/False)

3. _____ contacts are controlled directly by the timer coil, as in a general-purpose control relay.

(a) timed

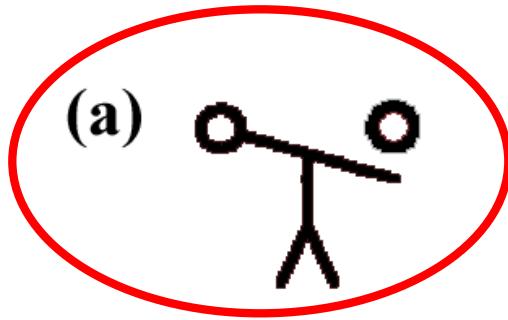
(b) NO

(c) instantaneous

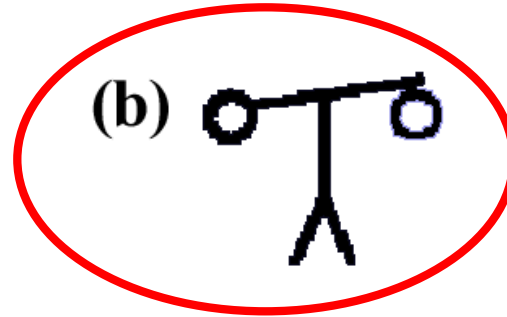
(d) NC

4. When a relay timer coil is de-energized, the timed contacts return instantaneously to their normal state. (True/False)

5. Which of the following symbols represents a normally open timed closed contact?



6. Which of the following symbols represents a normally closed timed open contact?

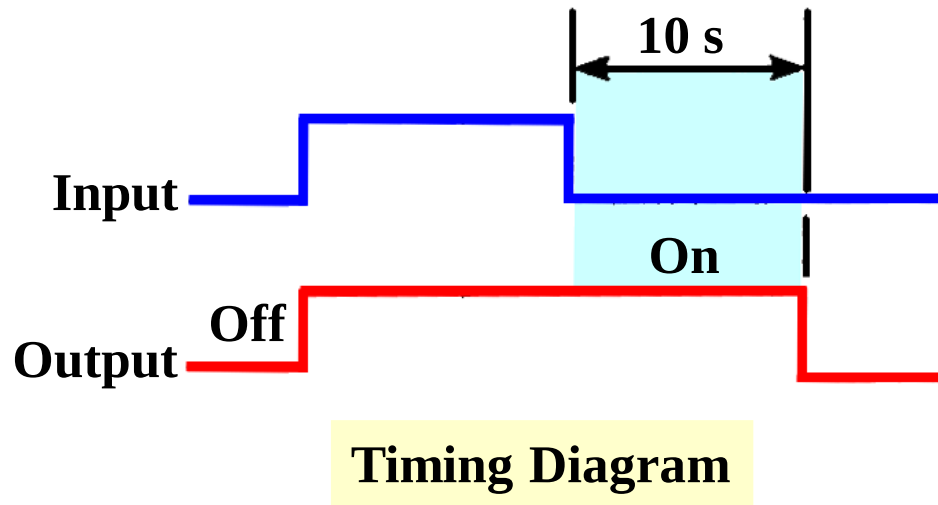


7. The timed relay contact shown is designed to operate so that:



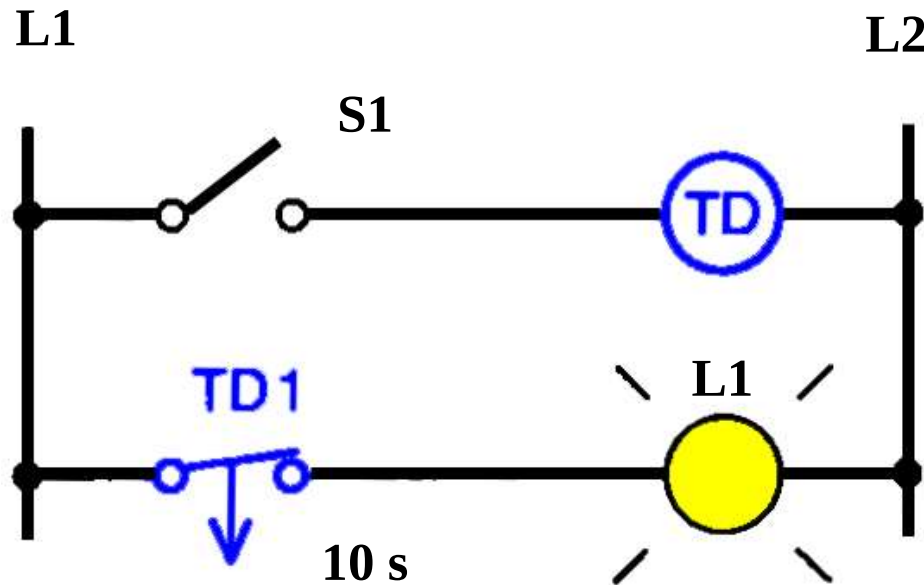
- a. when the relay coil is energized, there is a time delay in closing**
- b. when the relay coil is energized, there is a time delay in opening**
- c. when the relay coil is de-energized, there is a time delay in closing**
- d. when the relay coil is de-energized, there is a time delay in opening**

8. The timing diagram shown is that of an



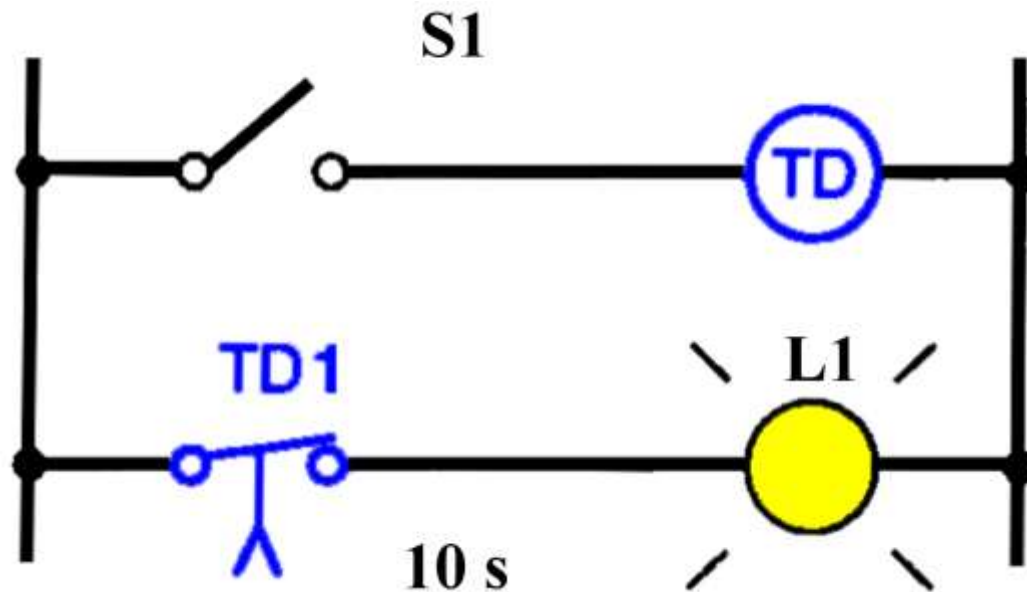
- a. on-delay timer circuit (NOTC contact)
- b. on-delay timer circuit (NCTO contact)
- c. off-delay timer circuit (NCTC contact)
- d. off-delay timer circuit (NOTO contact)

9. In the circuit shown, the light will stay *off*



- a. as long as S1 is closed
- b. for 10 seconds after coil TD is energized
- c. for 10 seconds after coil TD is de-energized
- d. both a and c

10. In the circuit shown, the light will stay *on*



- a. as long as S1 is closed
- b. as long as S1 is open
- c. for 10 seconds after S1 is closed
- d. both b and c

Programmed Timer Instructions

PLC timers are output instructions that provide the same functions as timing relays and solid state timers.

Some advantages of PLC timers:



- **their settings can be altered easily**
- **the number of PLC timers used can be increased or decreased by programming changes without wiring changes**
- **timer accuracy and repeatability are extremely high**

RSLogic Timer Commands



Command

Name

Description

TON

Timer On-Delay

Counts time base intervals when the instruction is “true”

RSLogic Timer Commands



Command	Name	Description
TOF	Timer Off-Delay	Counts time base intervals when the instruction is “false”

RSLogic Timer Commands



Command

Name

Description

RTO

**Retentive Timer
ON**

Counts time base intervals when the instruction is "true" and retains the accumulated value when the instruction goes "false" or when power cycle occurs

RES

Reset

When this instruction is "true" it resets the count of the RTO counter

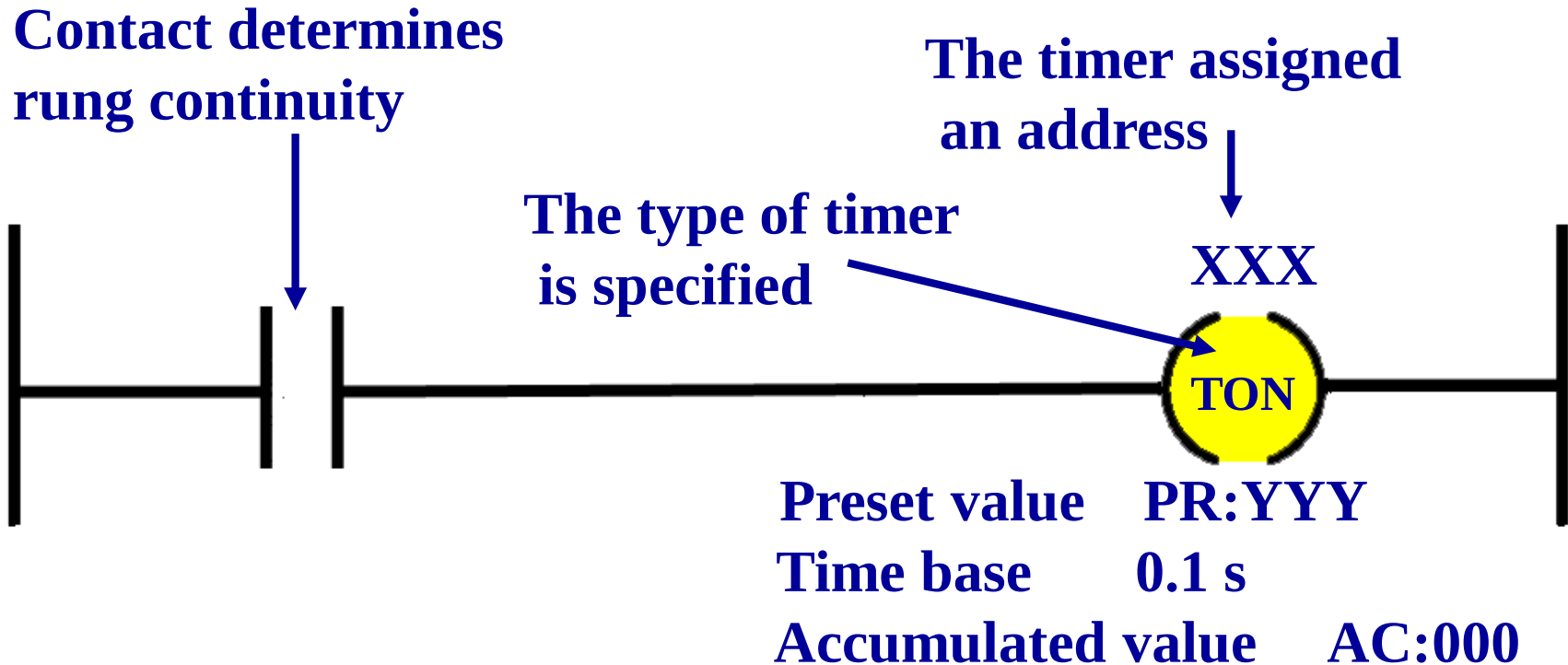
Quantities Associated with the Timer Instruction

Preset Time – Represents the time duration of the timing circuit. For example, if a time delay of 10 s is required, the timer will have a preset of 10 s.

Accumulated Time – Represents the amount of time that has elapsed from the moment the timing coil became energized.

Time Base – Timers can typically be programmed with several different time bases: 1 s, 0.1 s, and 0.01 s are typical time bases. For example, if you enter 0.1 for the time base and 50 for the preset time the timer would have a 5 s delay ($50 \times 0.1 \text{ s} = 5 \text{ s}$).

Coil-Formatted Timer Instruction

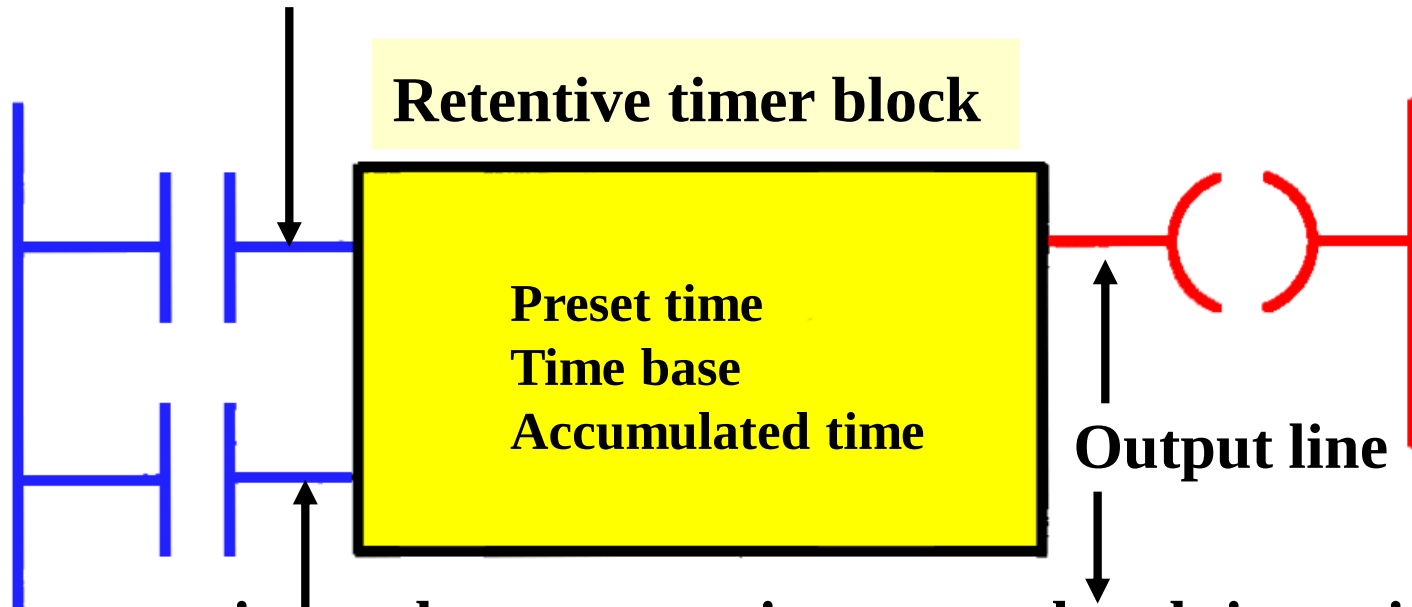


When the timer rung has logic continuity, the timer's accumulated value increases. When accumulated value equals the preset value, the output is energized and the timed output contact associated with the output is closed. The timed contact can be used as many times as you wish throughout the program as a NO or NC contact.

Generic Block-Formatted Timer Instruction

Timers are most often represented by boxes in a ladder logic.

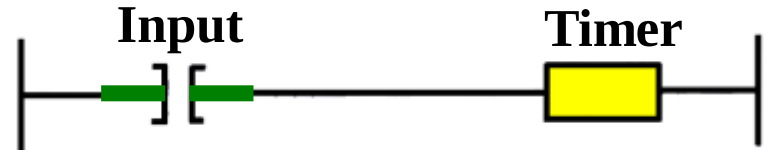
Control line controls the actual timing operation of the timer. Whenever this line is true the timer will time.



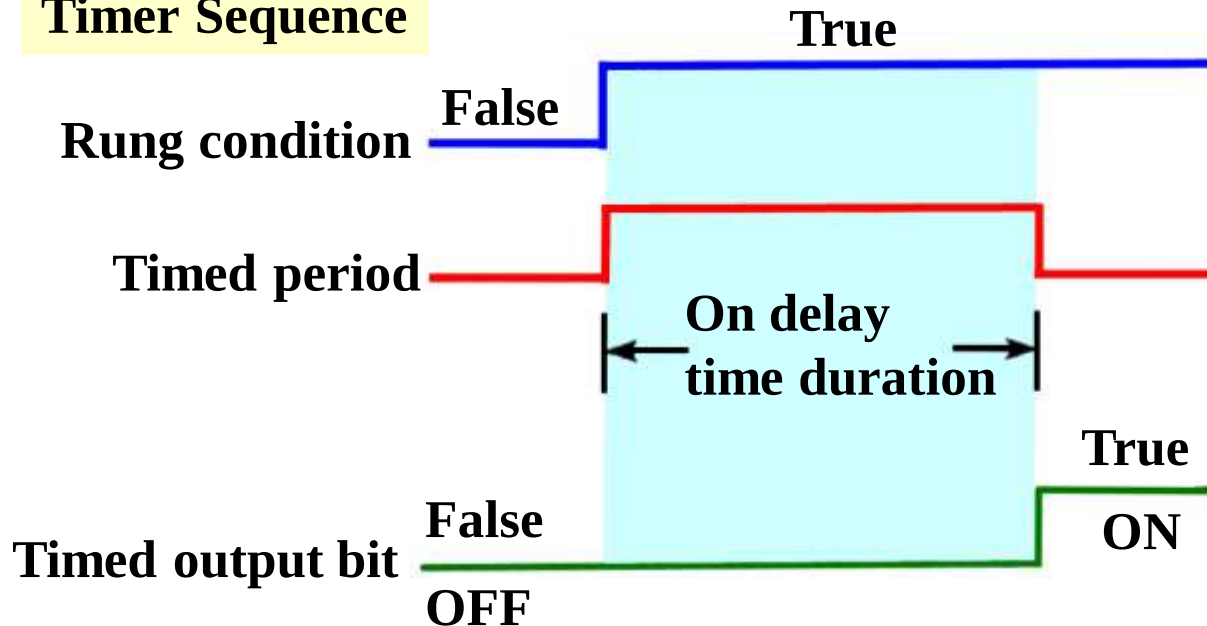
The timer continuously compares its accumulated time with its preset time. Its output is logic 0 as long as the accumulated time is less than the preset time. When the two become equal the output changes to logic 1.

On-Delay Timer Instruction

The *on-delay timer* operates so that, when the rung containing the timer is true, the timer time-out period commences.



Timer Sequence

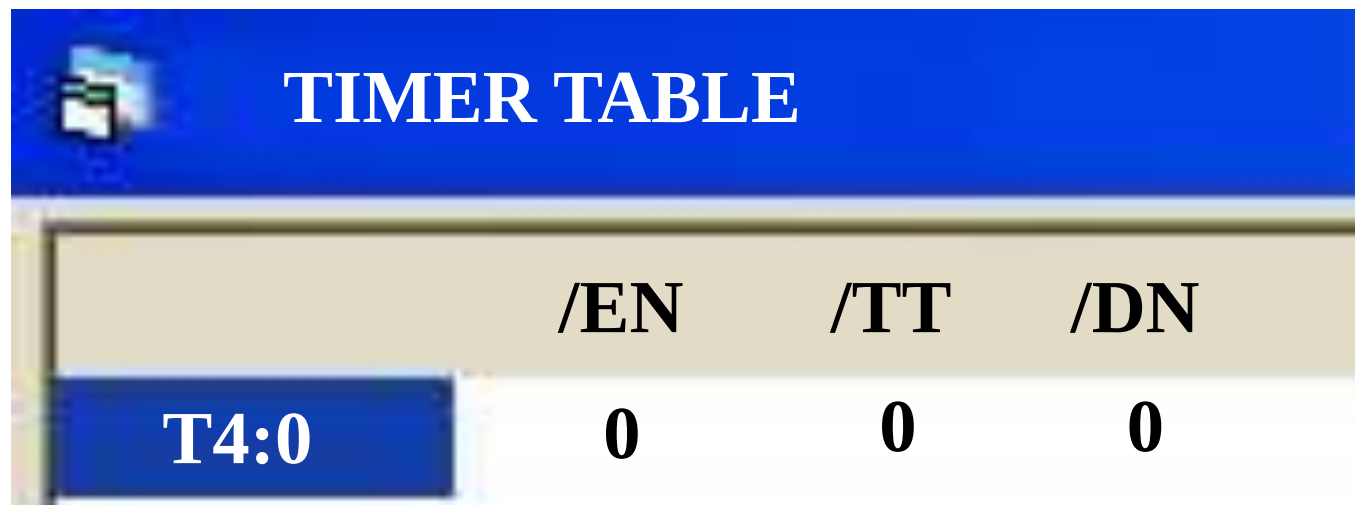


The timed output becomes true sometimes after the timer rung becomes true; hence the timer is said to have an on delay.

Allen-Bradley On-Delay Timer Instruction

Allen-Bradley PLC-5 and SLC-500 controller timer elements each take three data table words: *the control word, preset word, and accumulated word.*

The control word uses three control bits: Enable (EN) bit, Timer-Timing (TT) bit, and Done-Bit (DN).

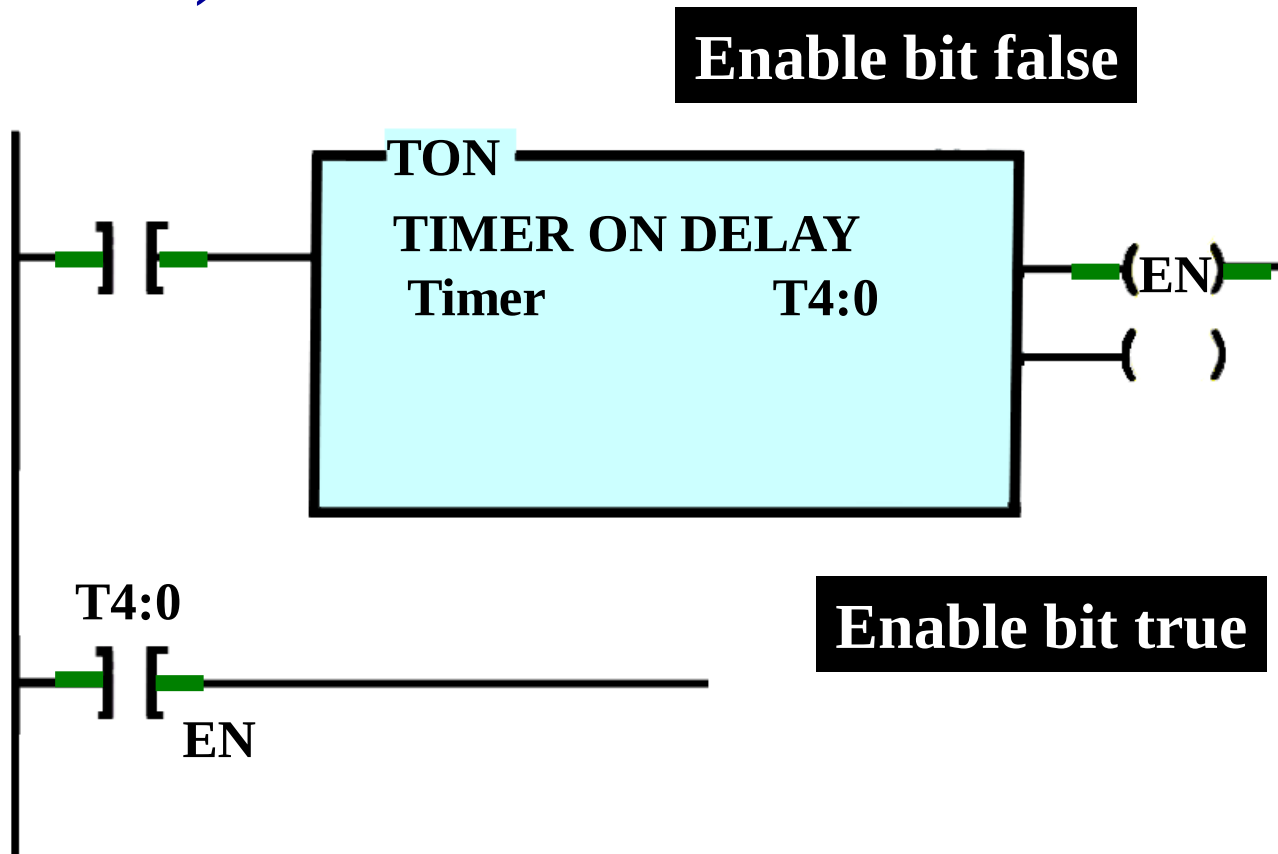


The image shows a screenshot of the Allen-Bradley ladder logic software interface. At the top, there is a blue header bar with a folder icon on the left and the text "TIMER TABLE" in white. Below this header is a table with a light beige background. The table has four columns: the first column is for the timer address, and the next three columns are for the control bits /EN, /TT, and /DN. The first row of the table shows the timer address "T4:0" in a blue box, and the values for the control bits are all "0".

	/EN	/TT	/DN
T4:0	0	0	0

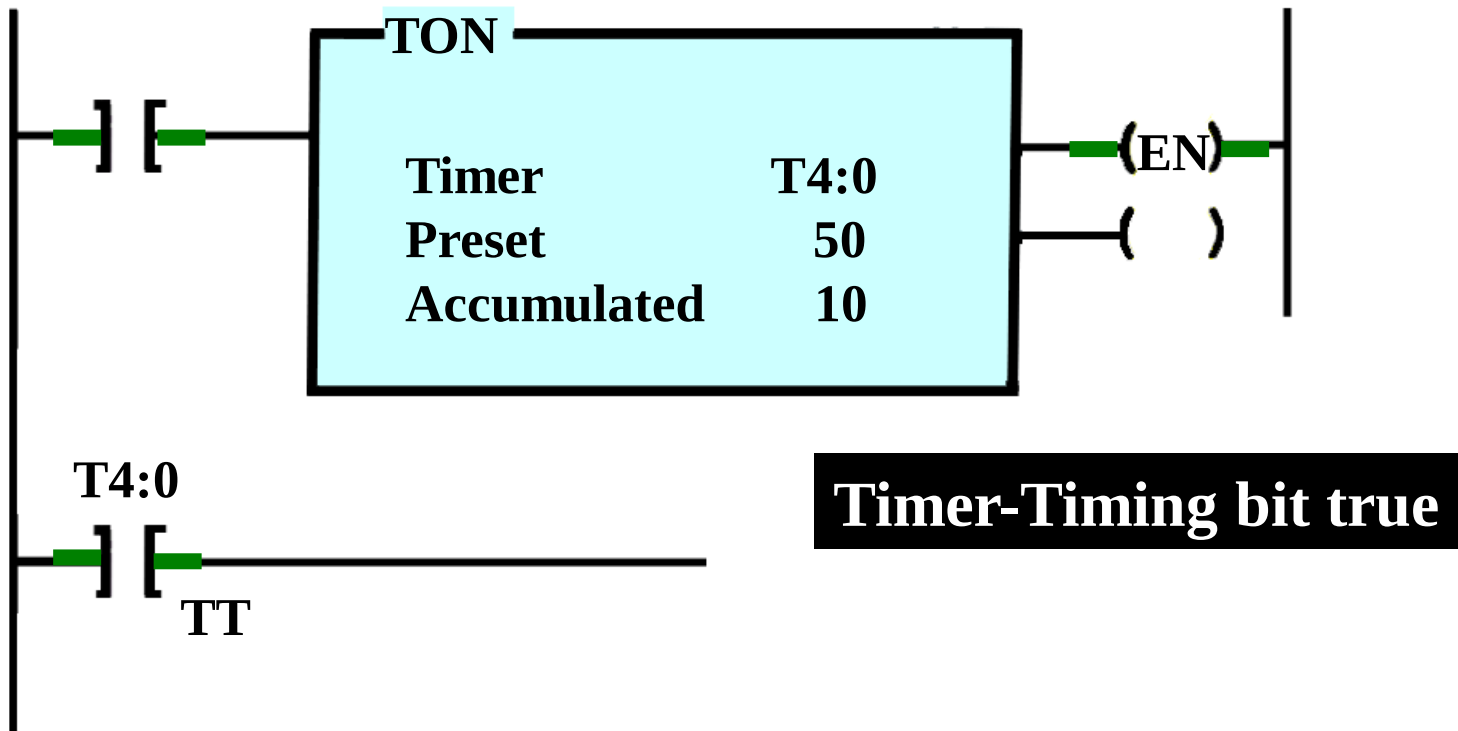
Allen-Bradley On-Delay Timer Instruction

The Enable (EN) bit is true (has a status of 1) whenever the timer instruction is true. When the timer instruction is false, the enable bit is false (has a status of 0)



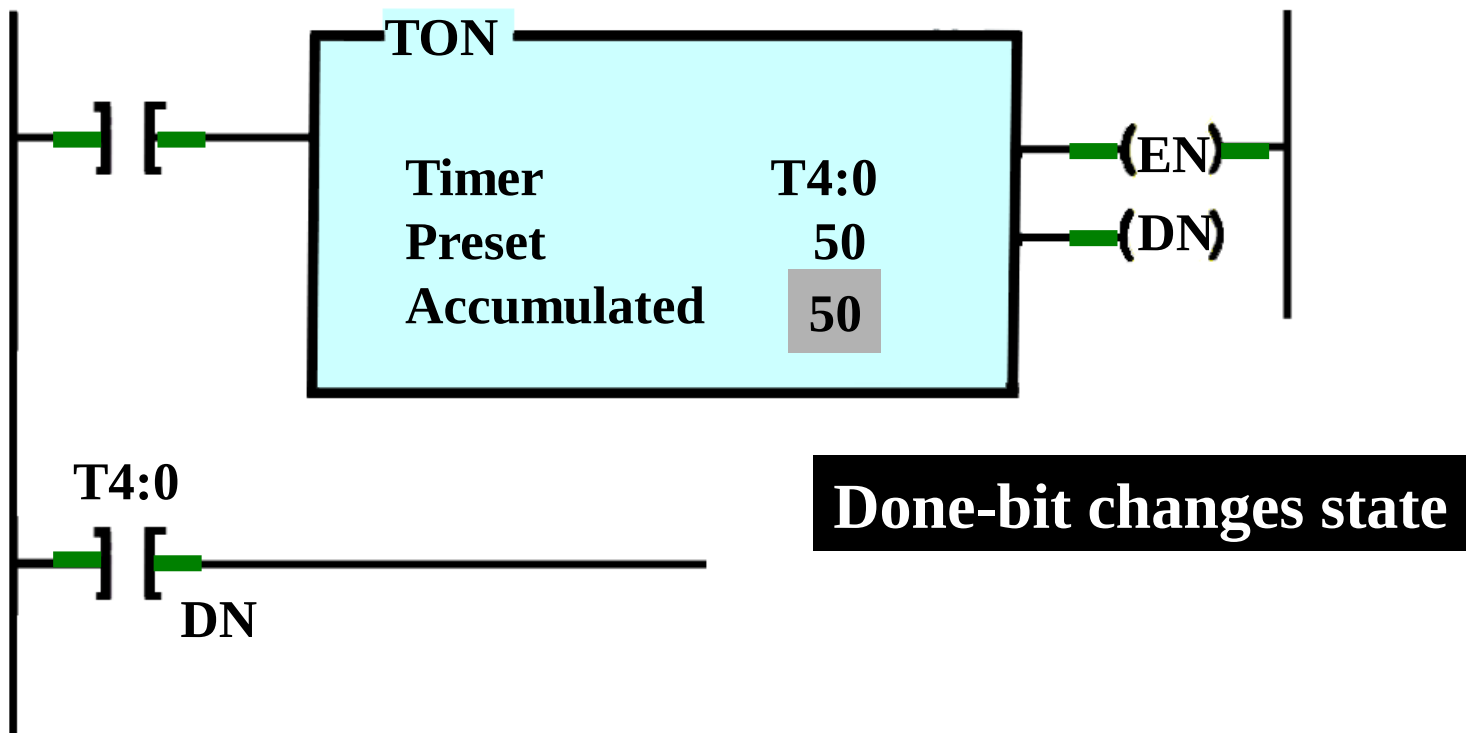
Allen-Bradley On-Delay Timer Instruction

The Timer-Timing (TT) bit is true whenever the accumulated value of the timer is changing, which means the timer is timing.



Allen-Bradley On-Delay Timer Instruction

The Done-Bit (DN) changes state whenever the accumulated value reaches the preset value. Its state depends on the type of timer being used.



Allen-Bradley On-Delay Timer Instruction

The preset value (*PRE*) word is the set point of the timer, that is, the value up to which the timer will time.

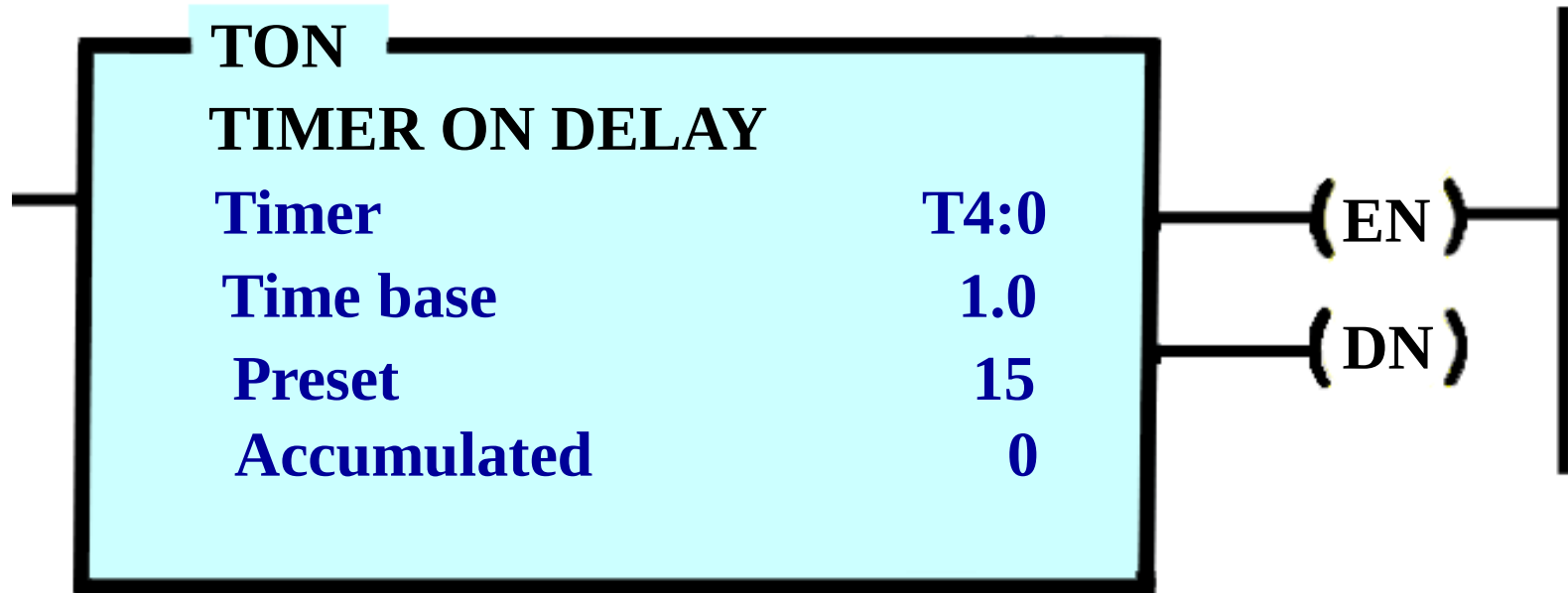
The accumulated value (*ACC*) word is the value that increments as the timer is timing. The accumulated value will stop incrementing when its value reaches the preset value.



TIMER TABLE					
	/EN	/TT	/DN	.PRE	.ACC
T4:0	0	0	0	0	0

Allen-Bradley On-Delay Timer Instruction

The information to be entered includes:



Timer number which must come from the timer file.

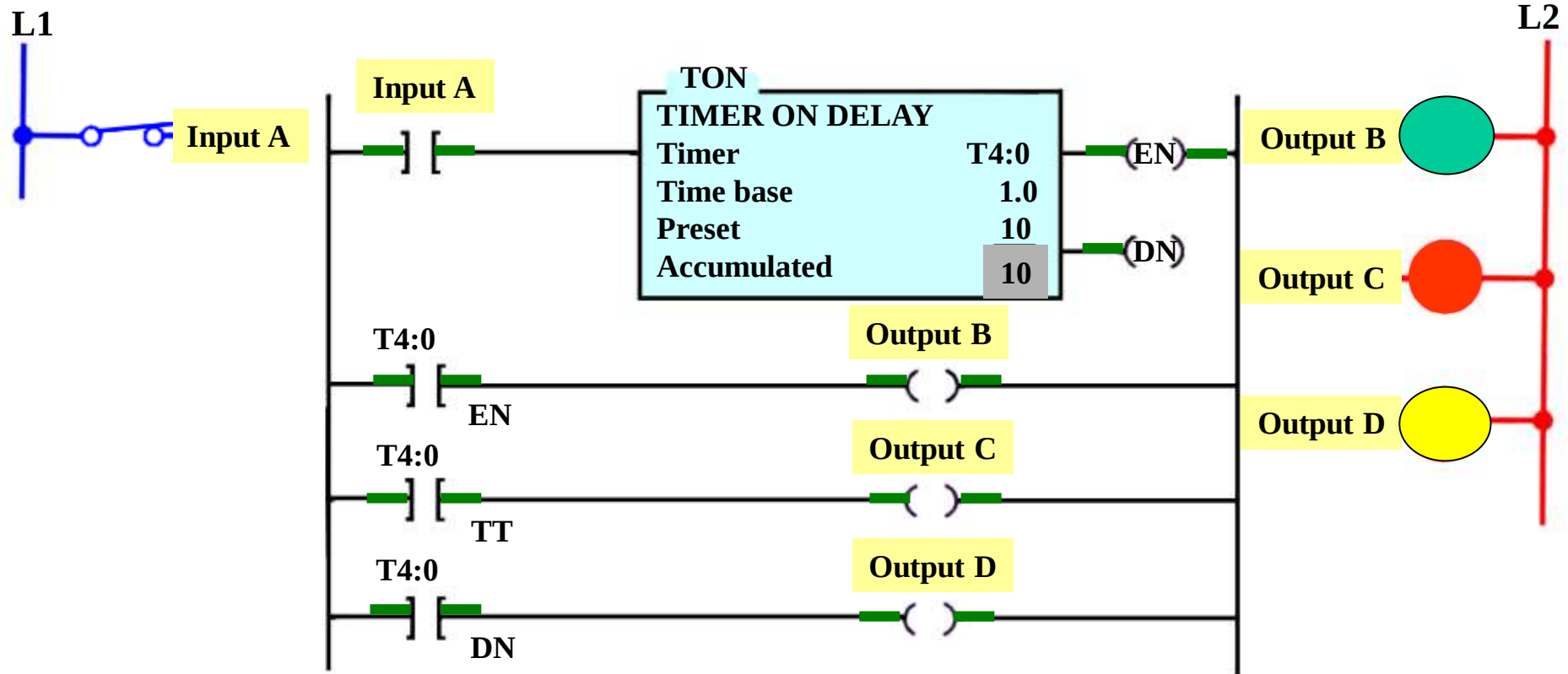
Time base which is expressed in seconds.

Preset value which is the length of the time delay.

Accumulated value which is normally entered as 0.

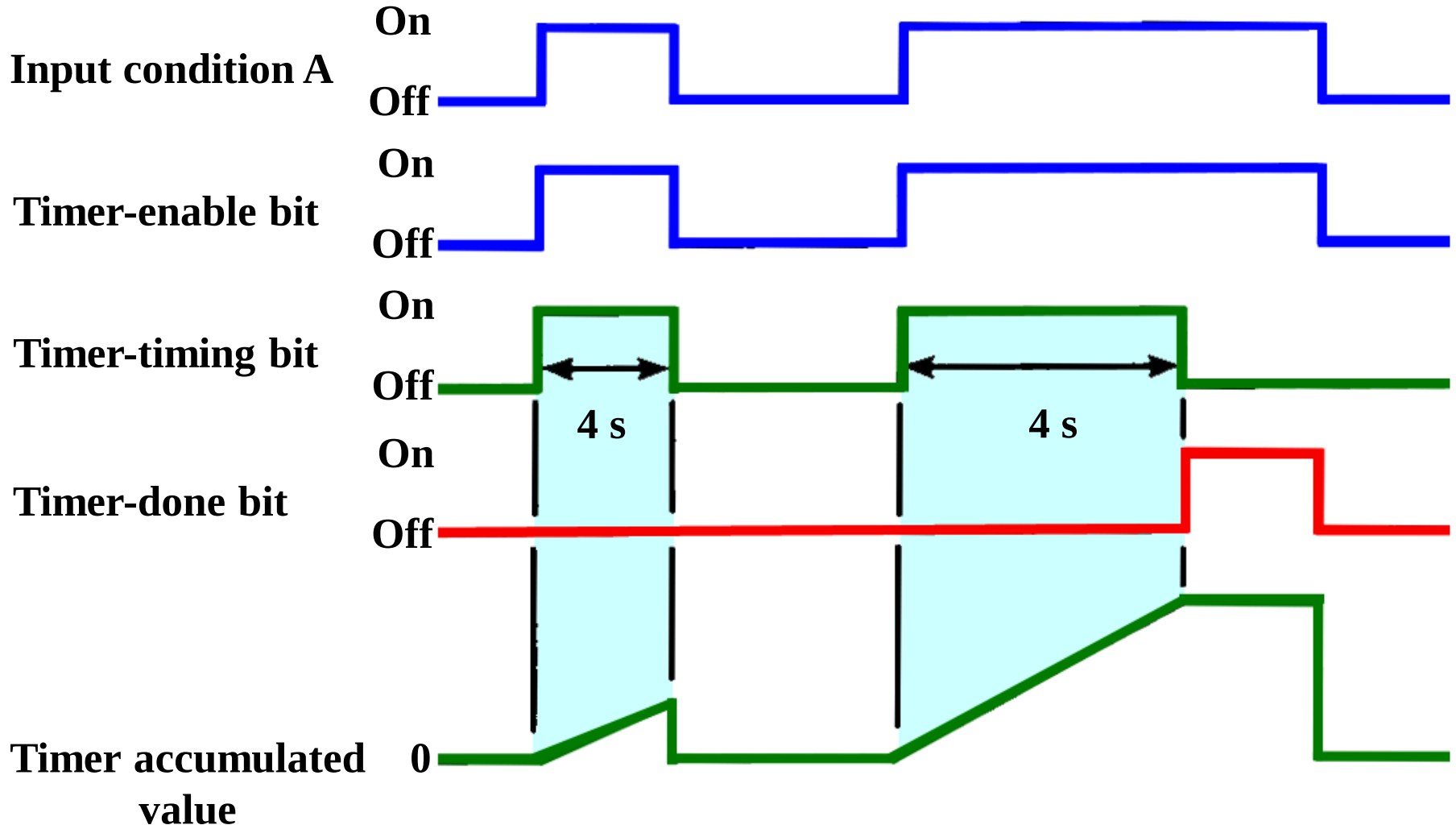
On-Delay Timer Program

Ladder Logic Program



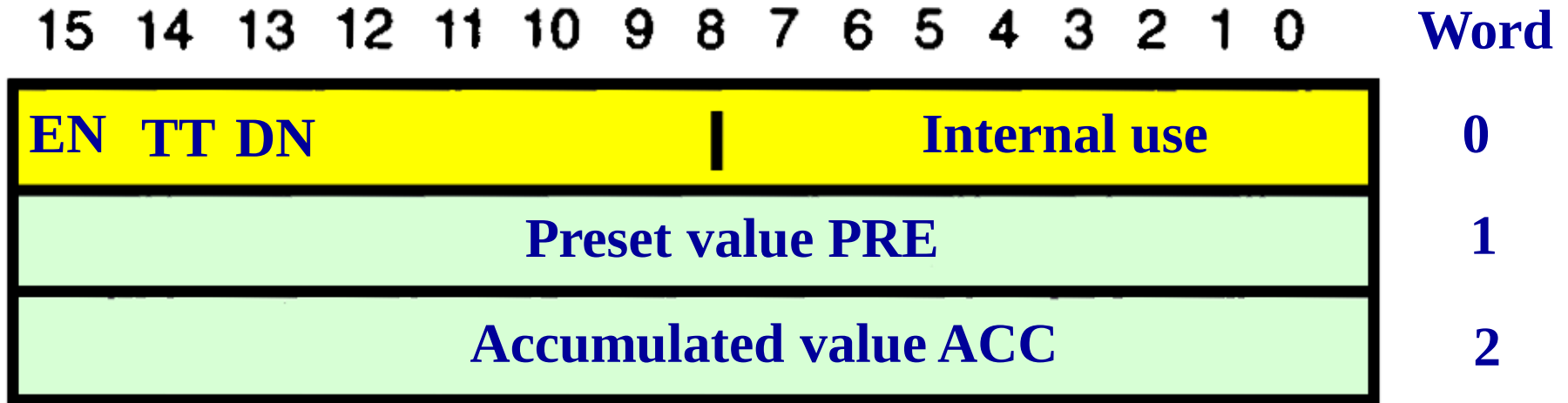
On-Delay Timer Program

Timing Diagram



On-Delay Timer Program

Timers are 3-word elements



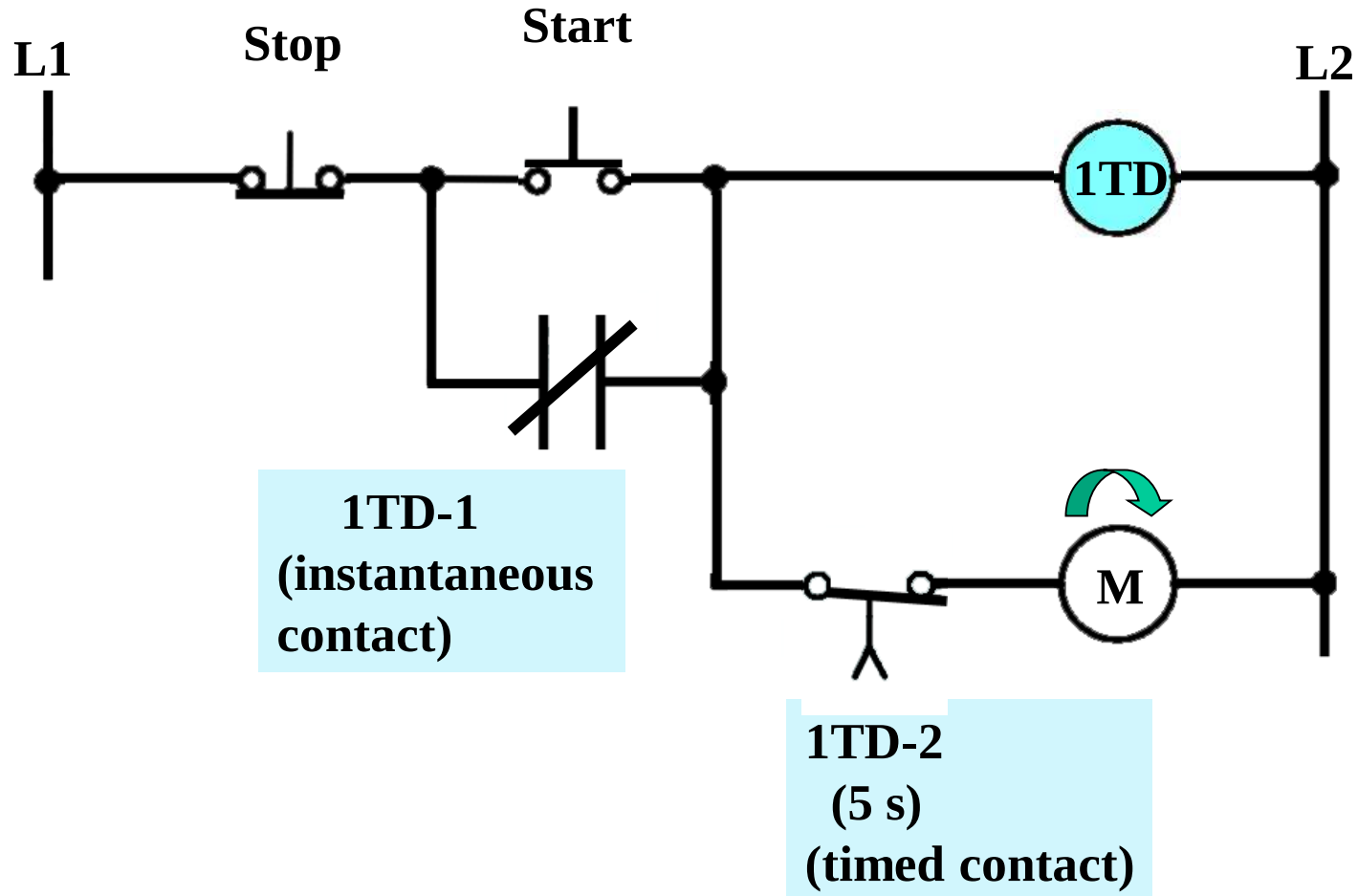
Word 0 is the control word

Word 1 stores the preset value

Word 2 stores the accumulated value

On-Delay Timer With Instantaneous Output

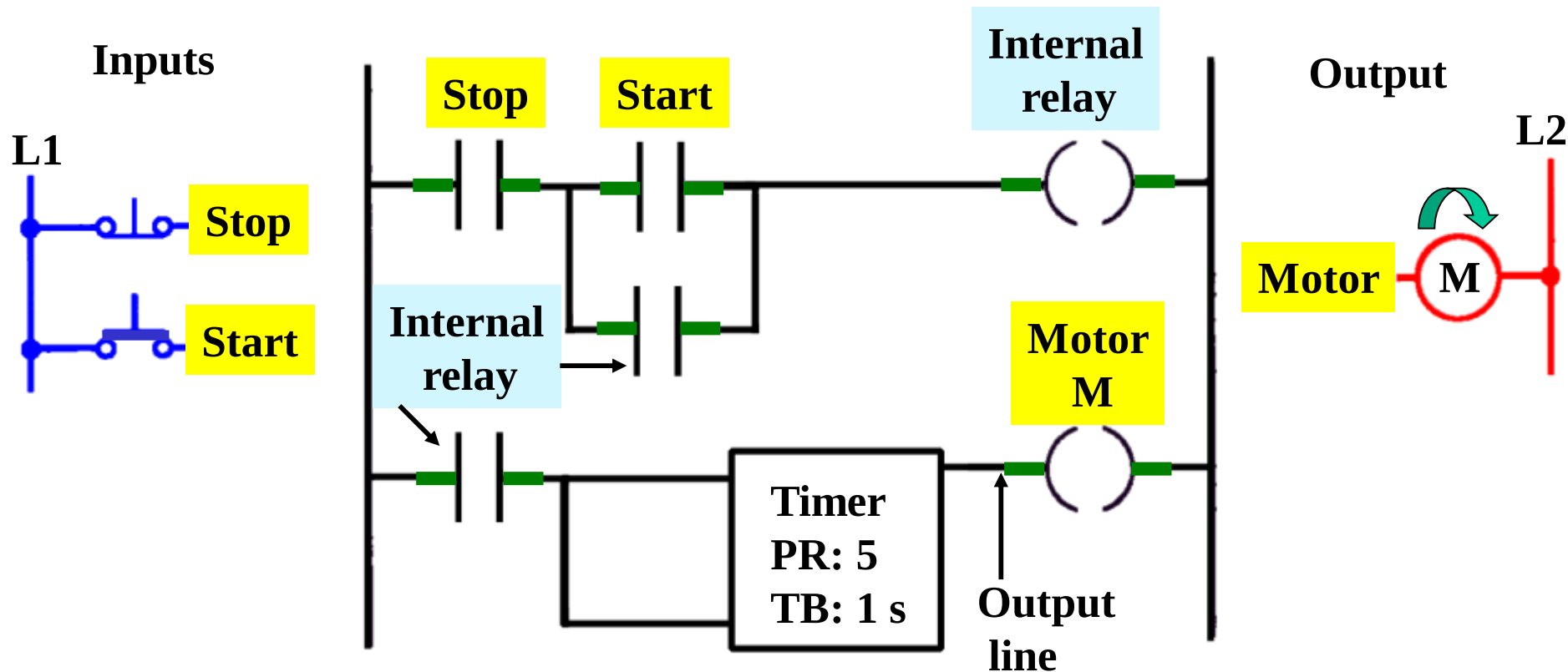
Relay Ladder Schematic Diagram



On-Delay Timer With Instantaneous Output

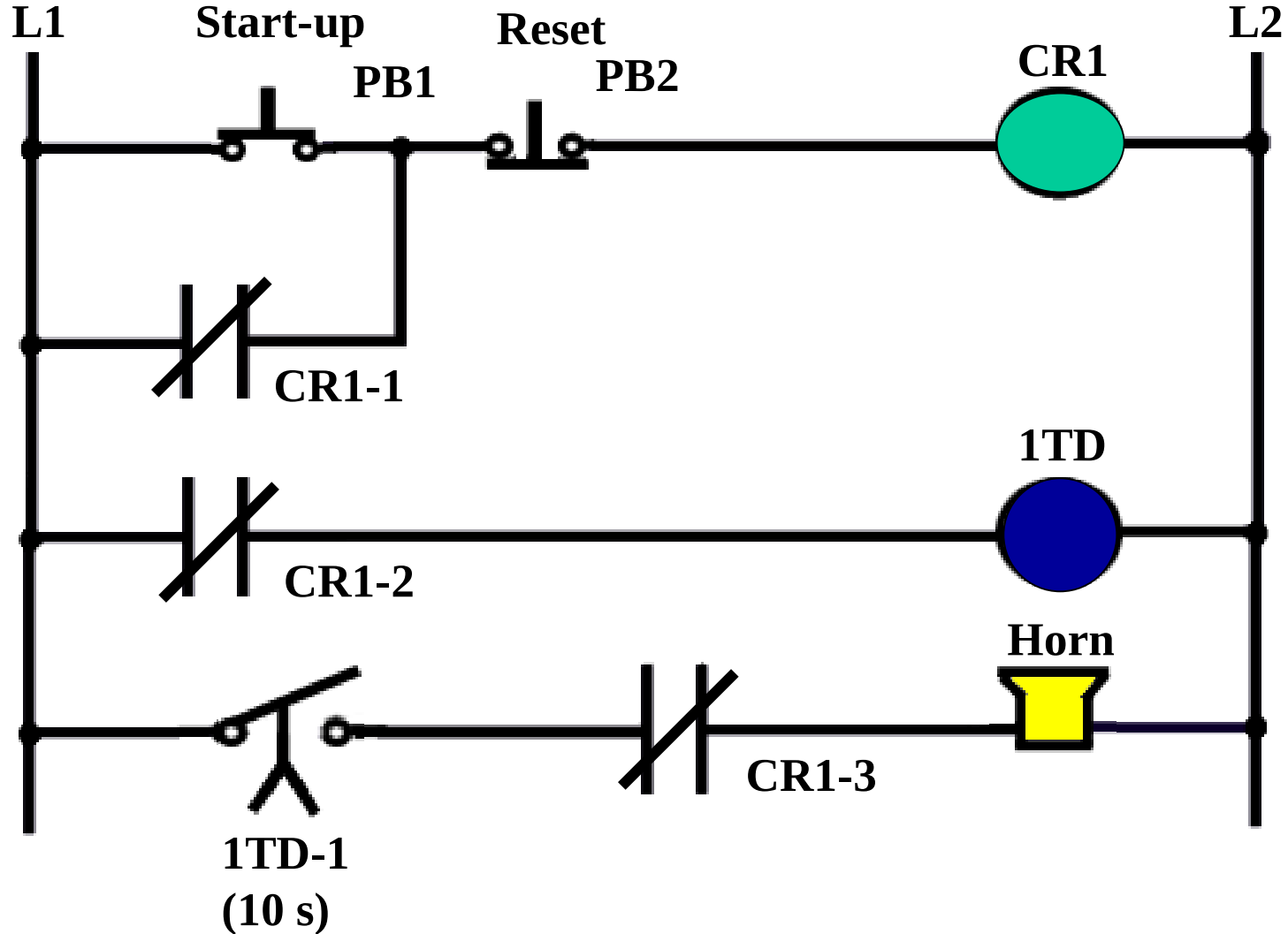
Programmed Circuit

Ladder logic program



Start-Up Warning Signal Circuit

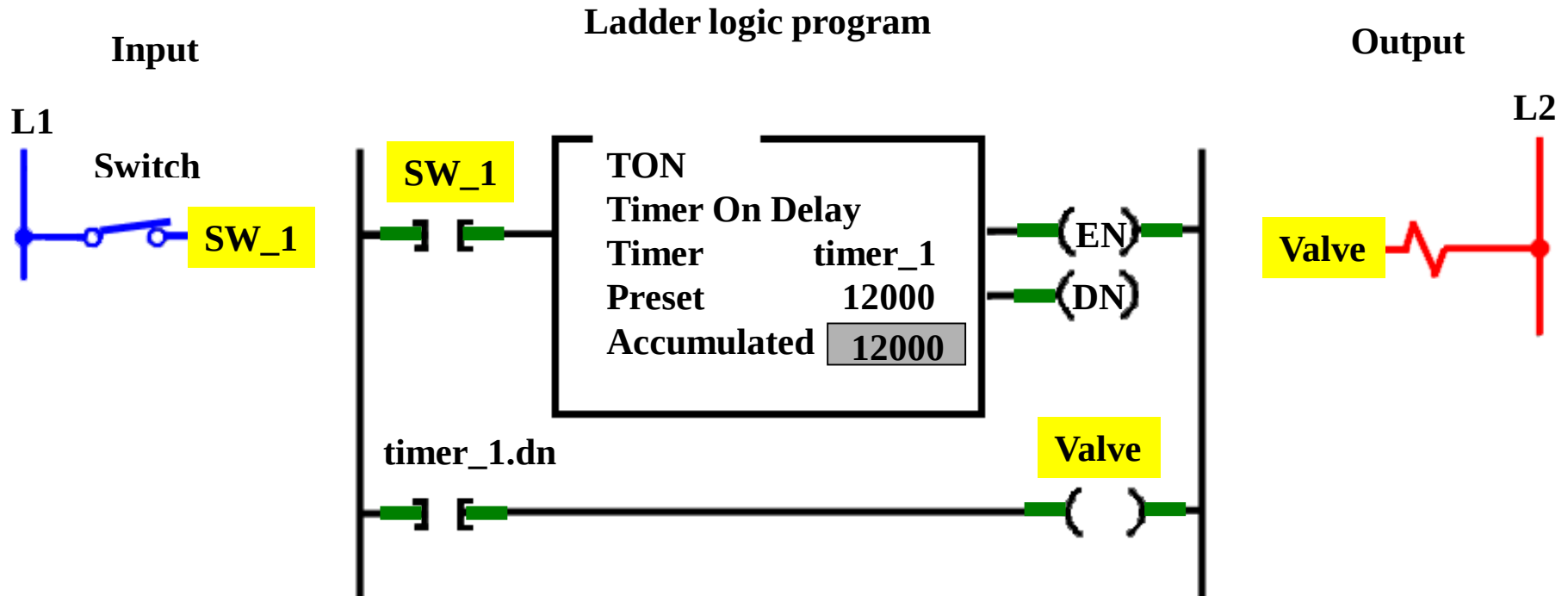
Relay Ladder Schematic Diagram



Programmed Circuit

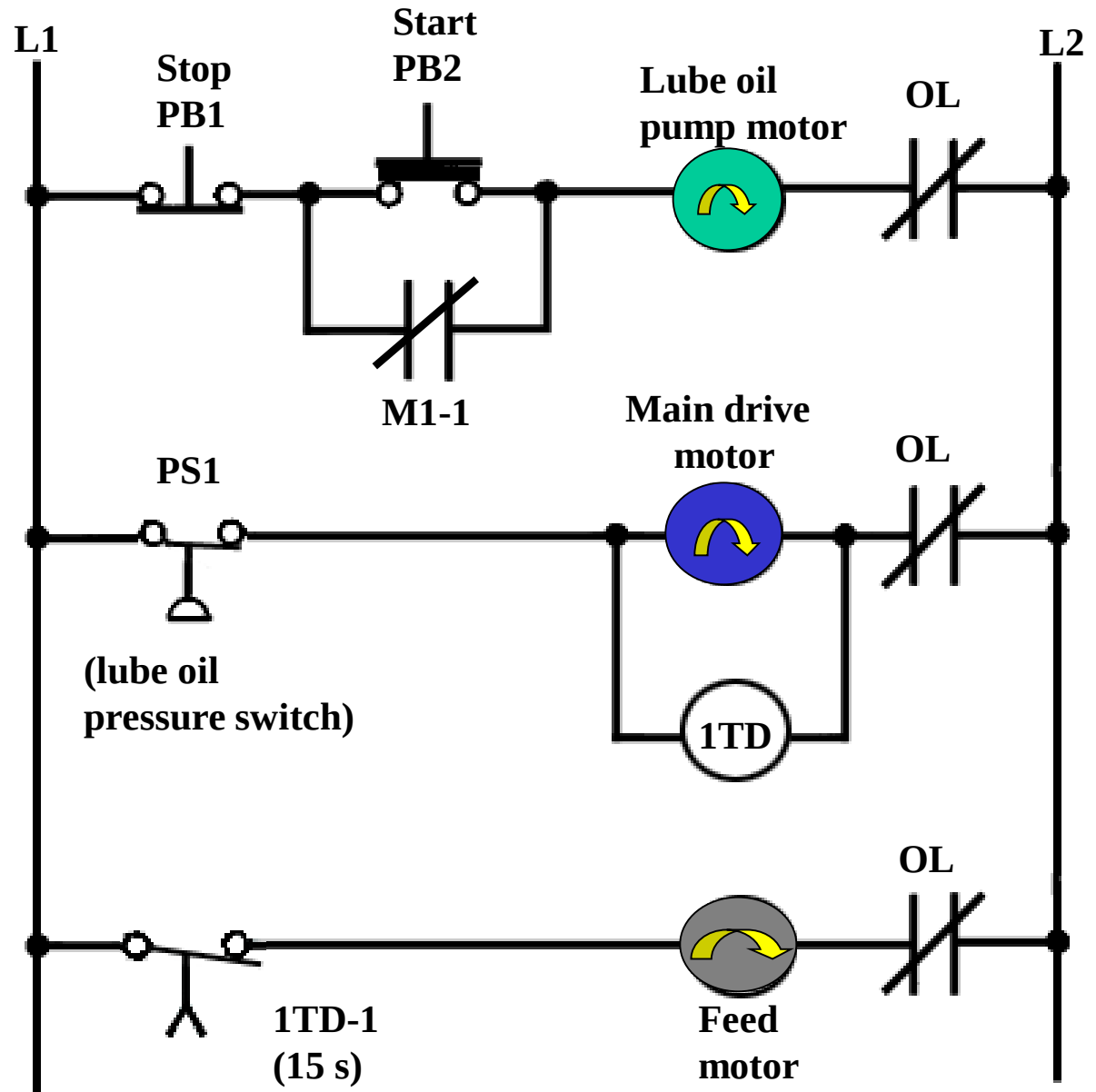


Timed Closed Solenoid Valve Program



Automatic Sequential Control System

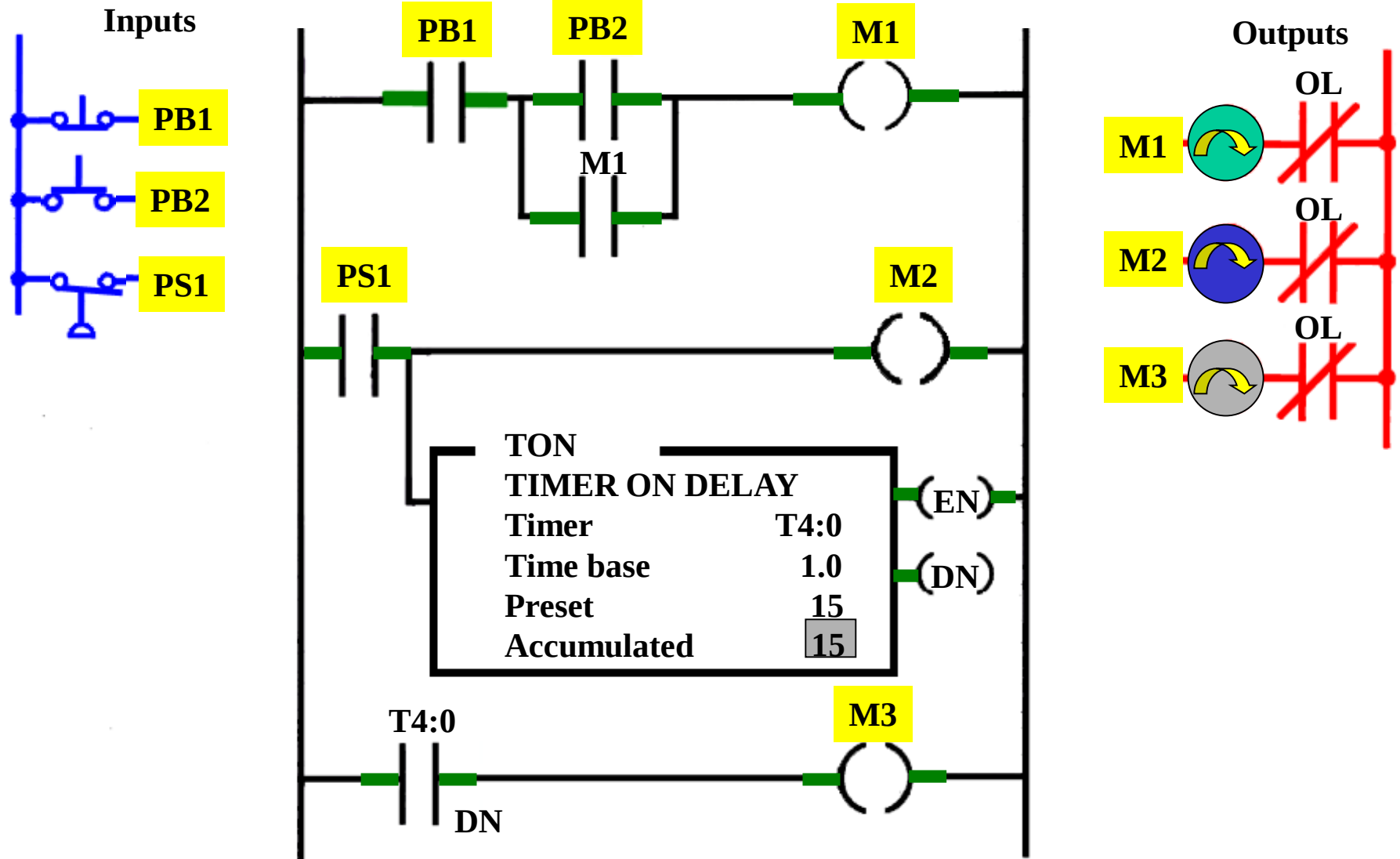
Relay Ladder Schematic Diagram



Automatic Sequential Control System

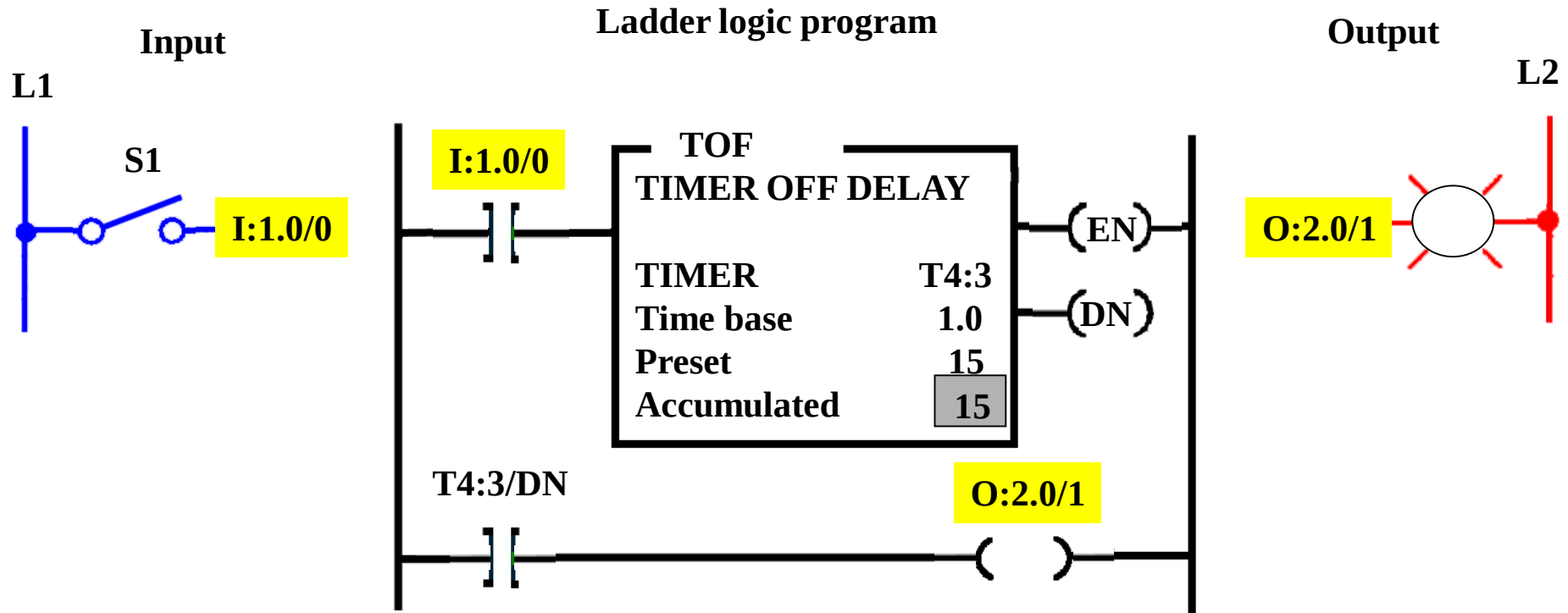
Programmed Circuit

Ladder logic program



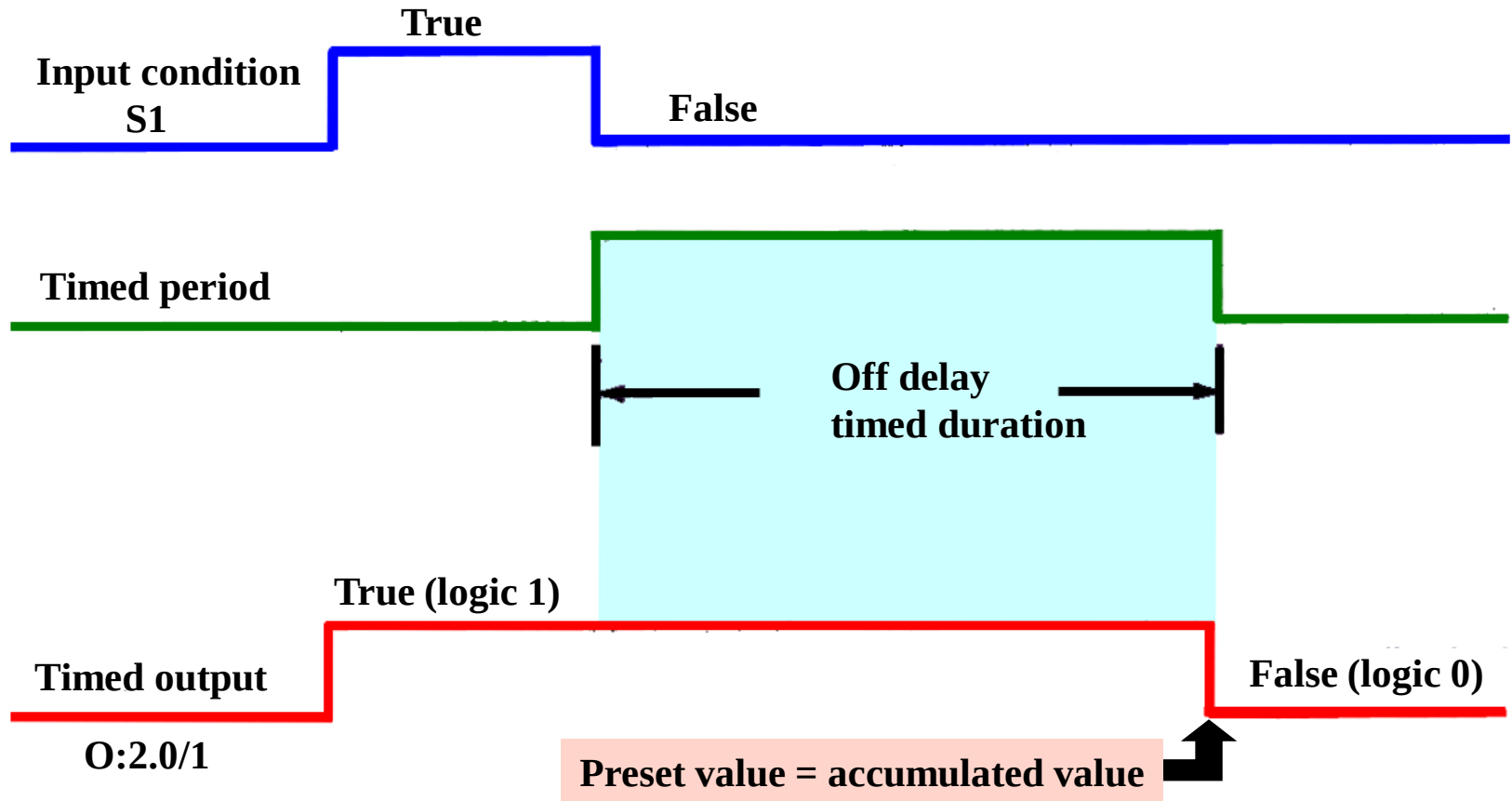
Off-Delay Programmed Timer

The *off-delay timer (TOF)* operation will keep the output energized for a period after the rung containing the timer has gone false.

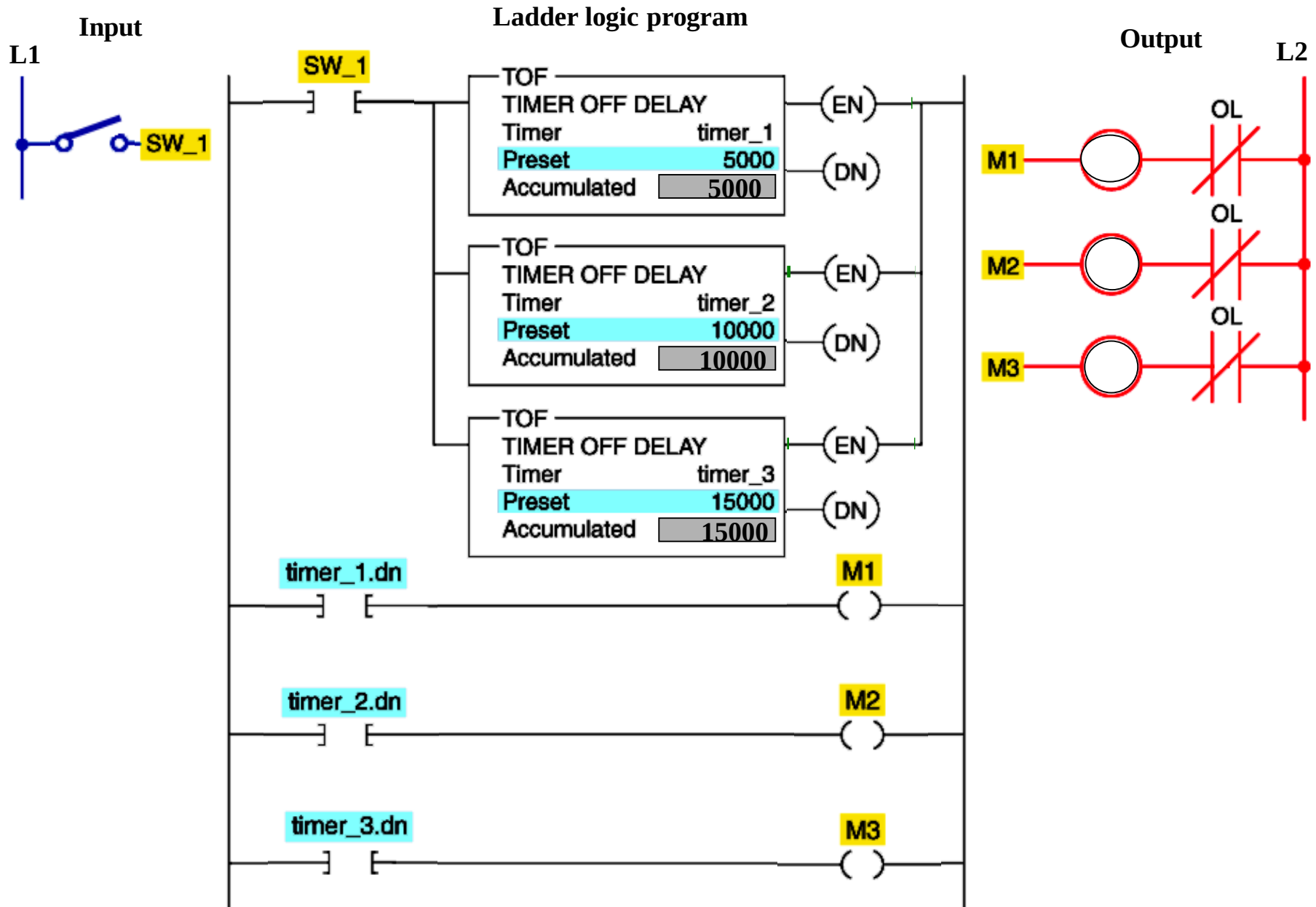


Off-Delay Programmed Timer

Timing Diagram

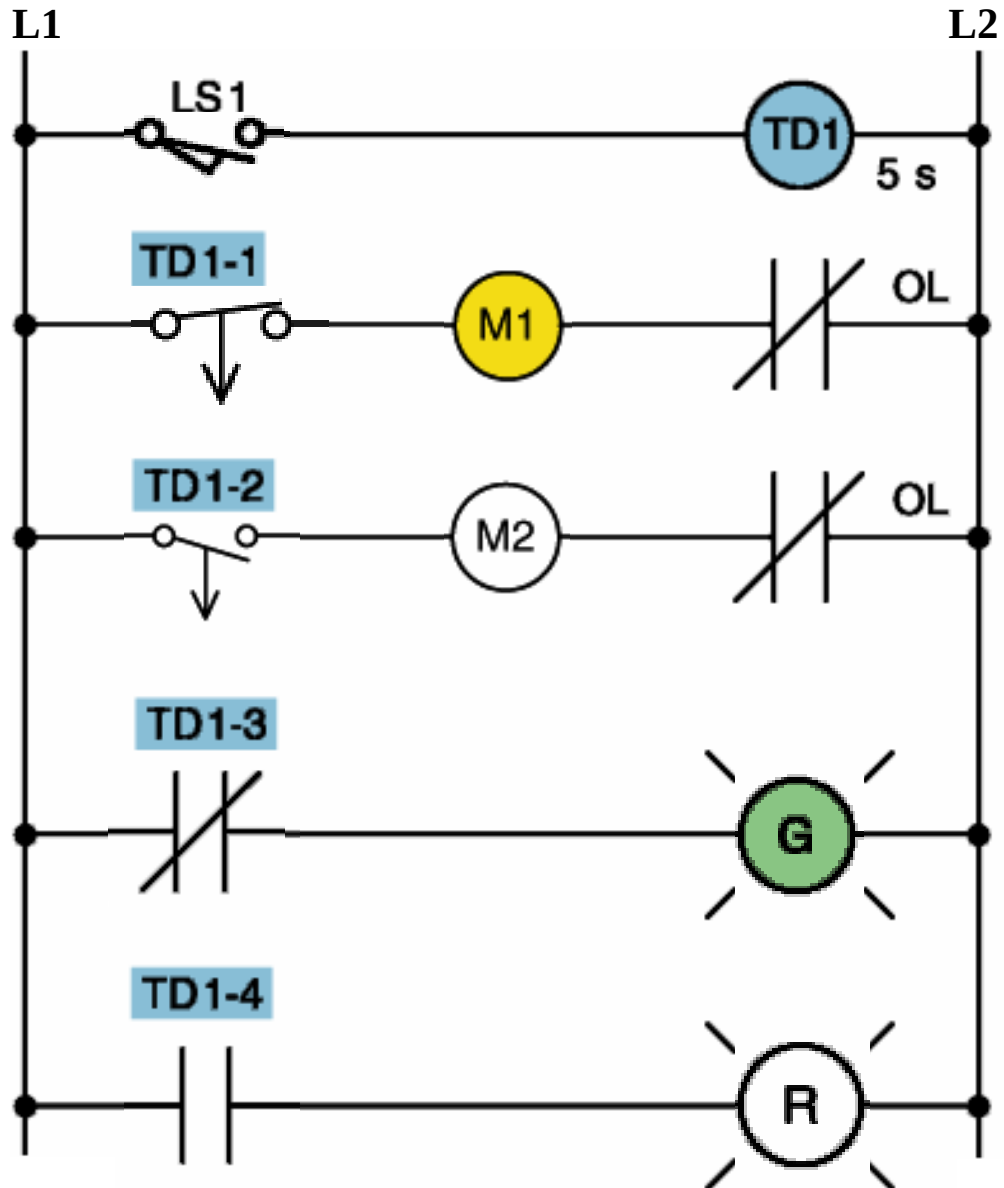


Off-Delay Timer Used To Switch Motors Off



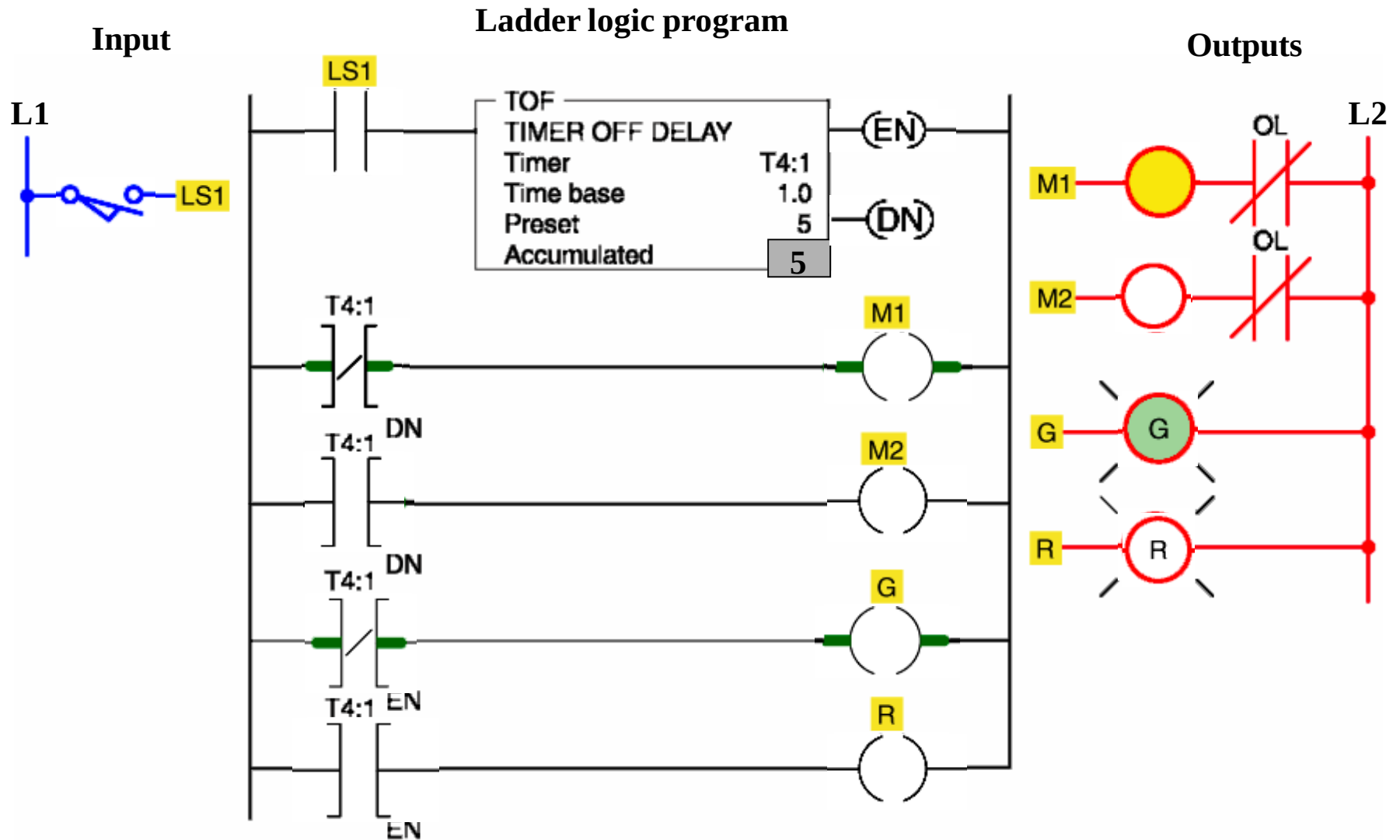
Pneumatic Off-Delay Timer

Relay Ladder Schematic Diagram



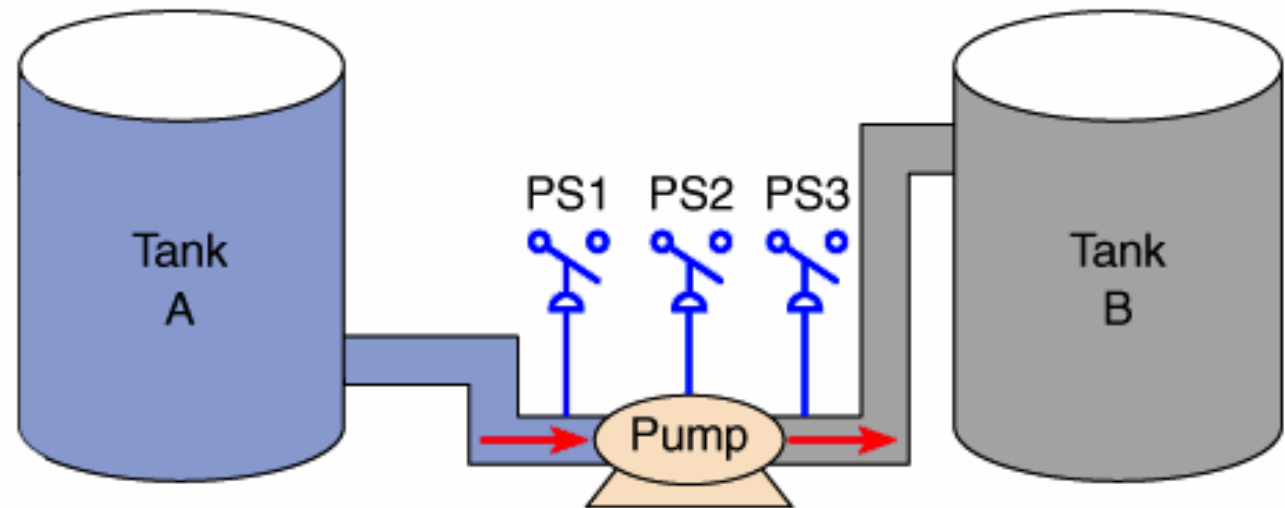
Programmed Pneumatic Off-Delay Timer

Equivalent Programmed Circuit



Fluid Pumping Process

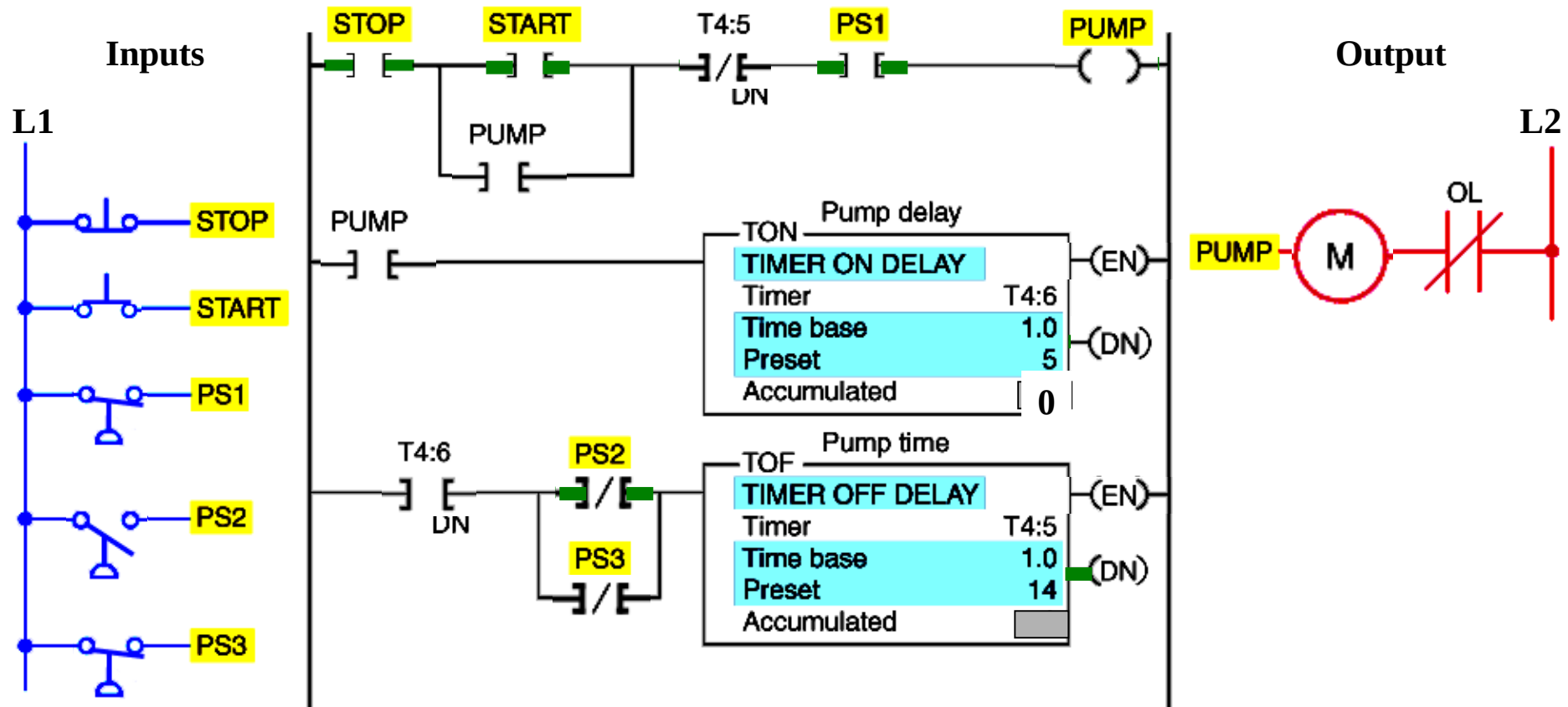
Operation



- Before starting, PS1 must be closed.
- When the pump start button is pressed, the pump starts. The button can then be released and the pump continues to operate.
- When the stop button is pushed, the pump stops.
- PS2 and PS3 must be closed for 5 s after the pump starts. If either PS2 or PS3 opens, the pump will shut off and will not be able to start again for another 14 s.

Fluid Pumping Process Program

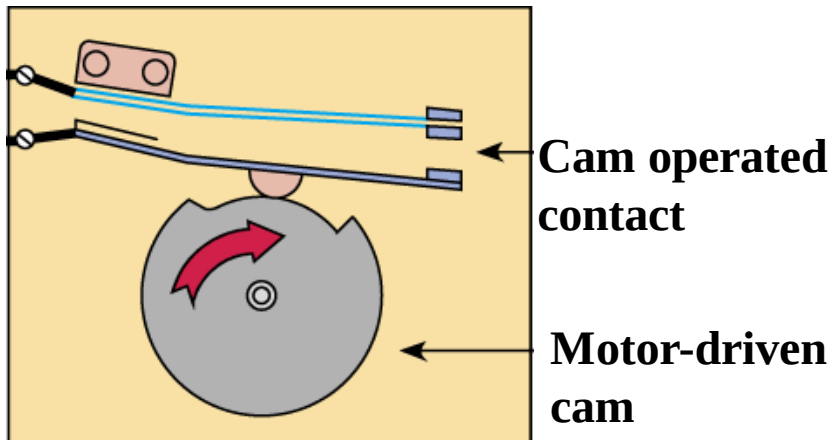
Ladder logic program



Retentive Timer

*A retentive timer accumulates time whenever the device receives power, and maintains the current time should power be removed from the device. Once the device accumulates time equal to its preset value, the contacts of the device change state. The retentive timer must be *intentionally reset* with a separate signal for the accumulated time to be reset.*

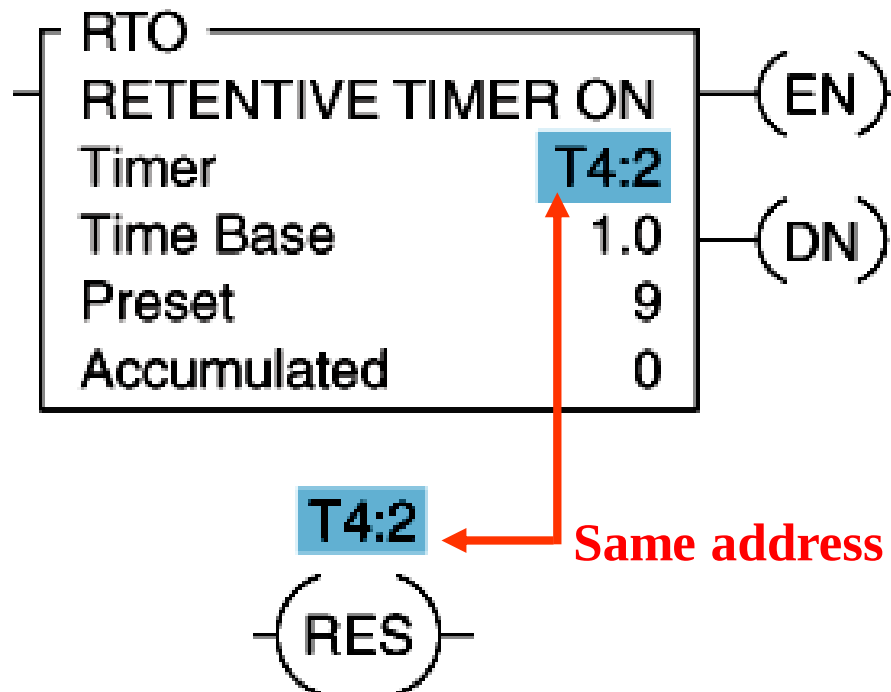
Electromechanical Retentive Timer



Once power is applied, the motor starts turning the cam. The positioning of the lobes determines the time it takes to activate the contacts. If power is removed from the motor, the shaft stops but *does not* reset.

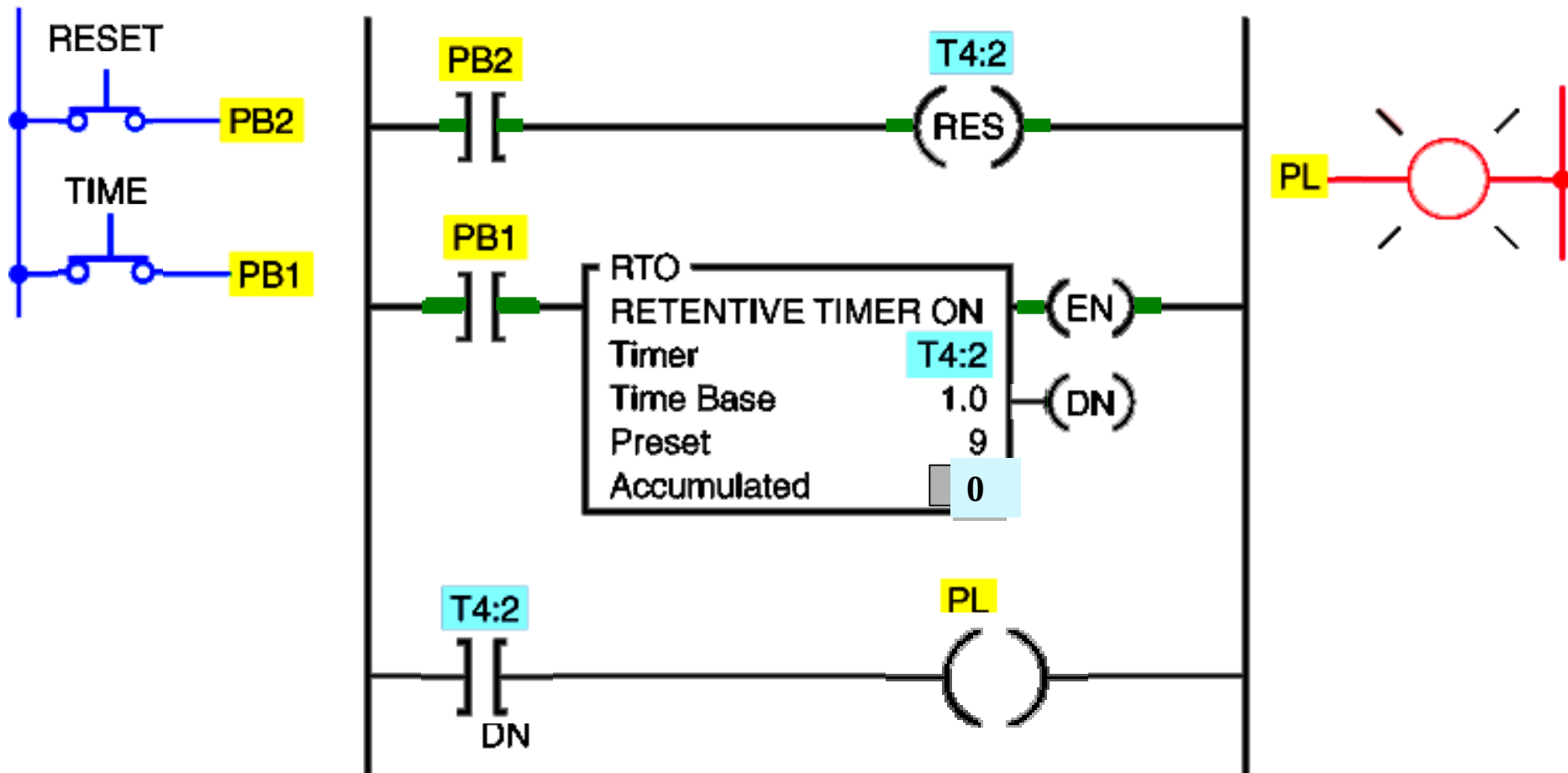
Retentive On-Delay Timer Program

The PLC-programmed **RETENTIVE ON-DELAY** timer (RTO) operates in the same way as the nonretentive on-delay timer (TON), with one major exception. This is a **retentive timer reset (RTR)** instruction.

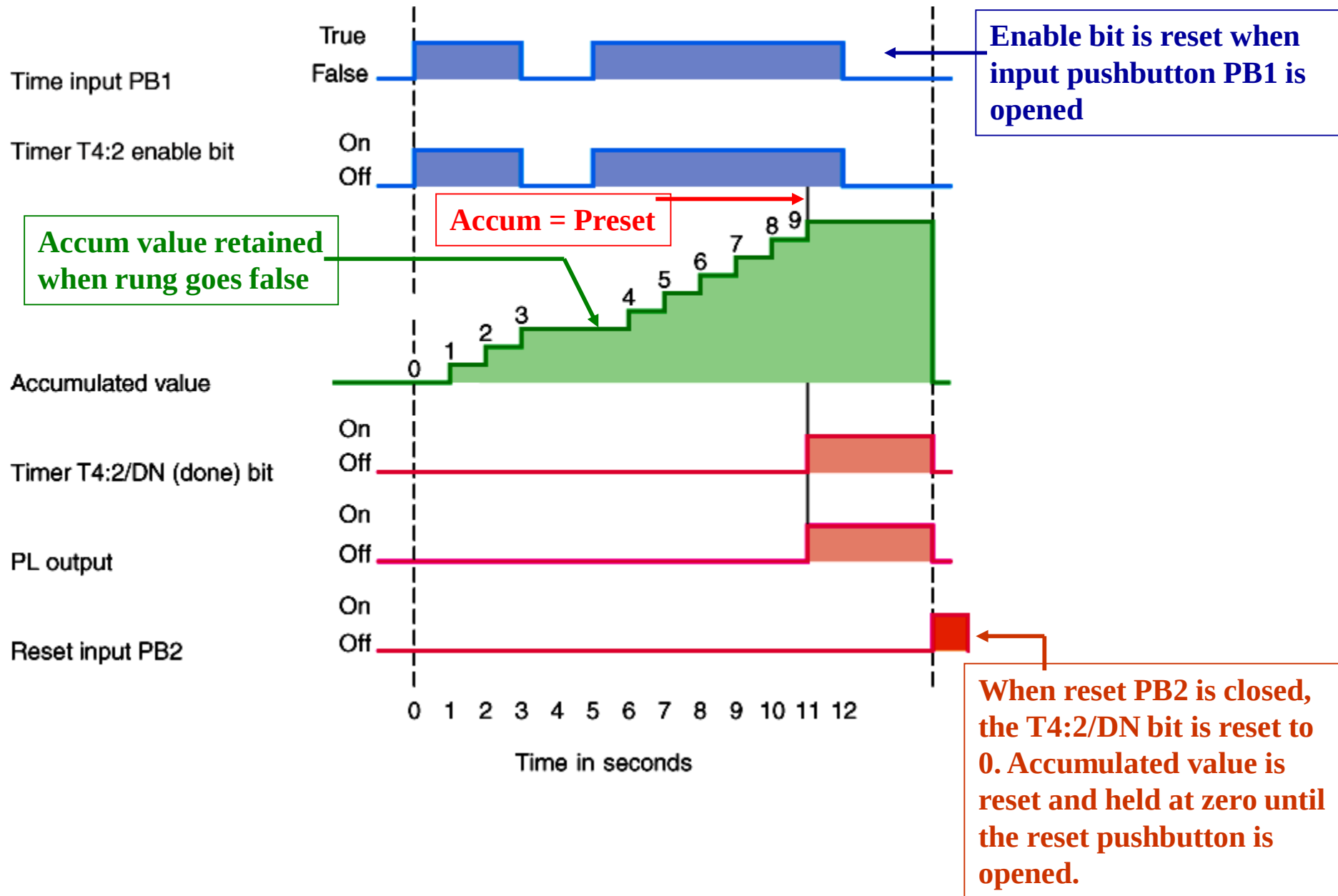


Unlike the TON, the RTO will hold its accumulated value when the timer rung goes false and will continue timing where it left off when the timer rung goes true again. This timer must be accompanied by a timer reset (RES) instruction to reset the accumulated value of the timer to zero.

Retentive On-Delay Timer Program

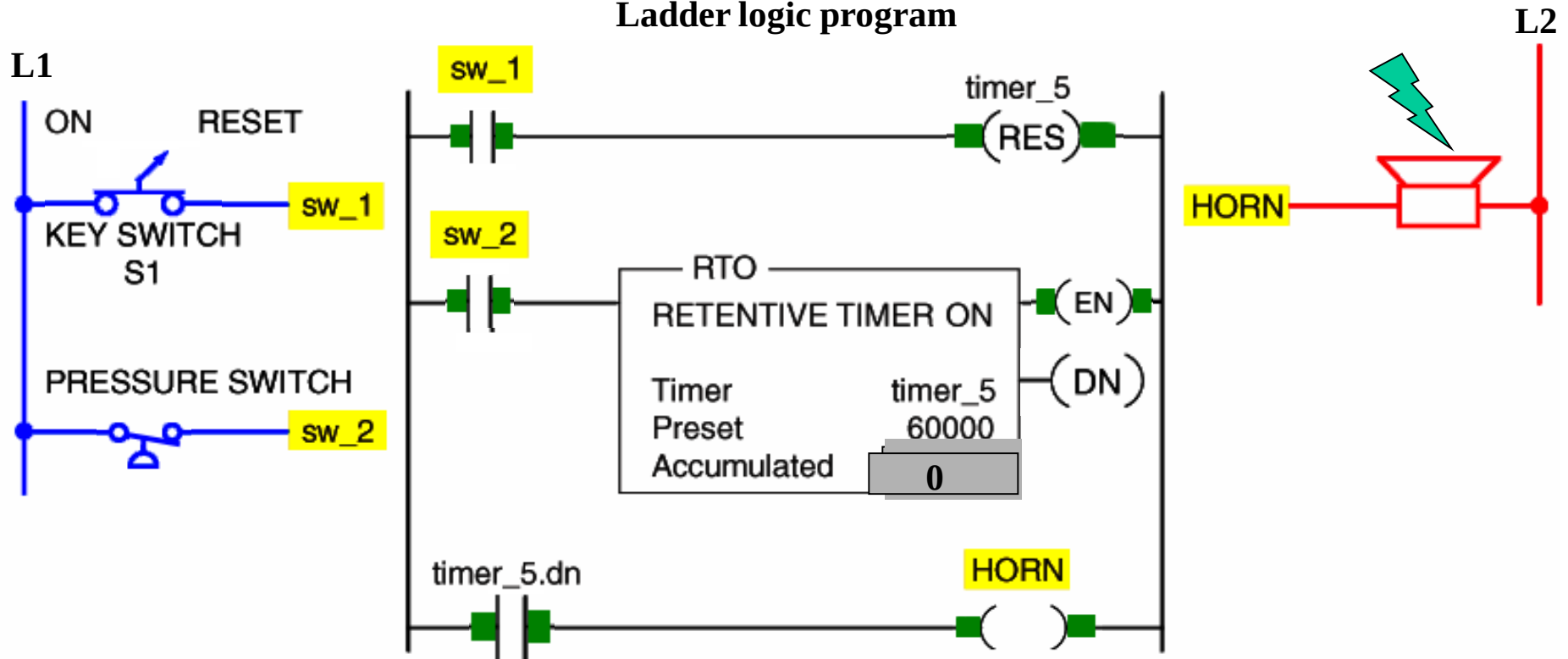


Retentive On-Delay Timing Chart



Retentive On-delay Alarm Program

Ladder logic program



The purpose of the RTO timer is to detect whenever a piping system has sustained a cumulative overpressure condition of 60 s. At that point, a horn is sounded automatically. You can silence the alarm by switching the key switch to the rest position.

Bearing Lubrication Program

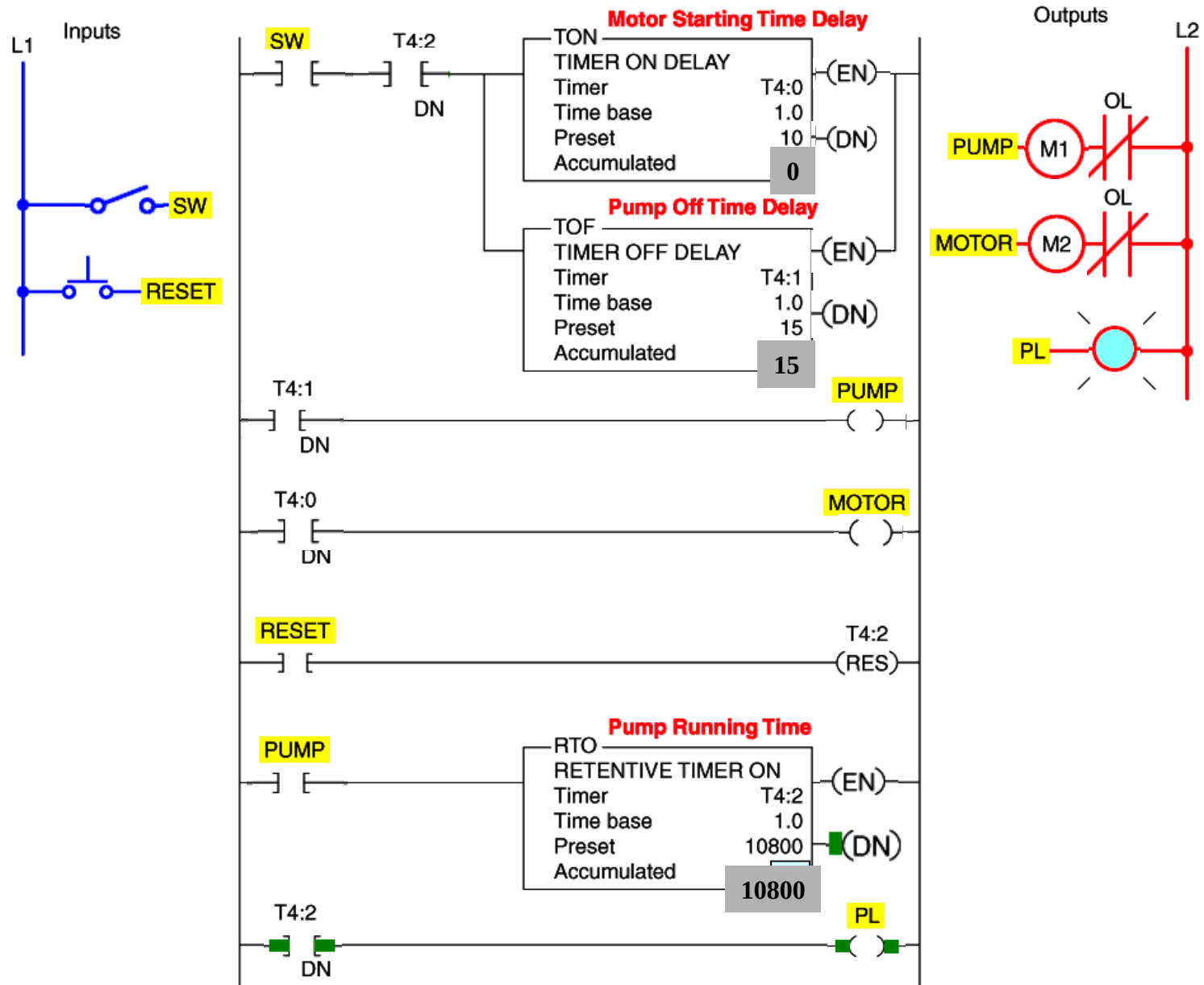
Sequence Of Operation



- To start the machine, the operator turns SW on.
- Before the motor shaft starts to turn, the bearings are supplied with oil by the pump for 10 s.
- The bearings also receive oil when the machine is running.
- When the operator turns SW off to stop the machine, the oil pump continues to supply oil for 15 s.
- A retentive timer is used to track the total running time of the pump. When the total running time is 3 h, the motor is shut down and a pilot light is turned on to indicate that the filter and oil need to be changed.
- A reset button is provided to reset the process after the filter and oil have been changed.

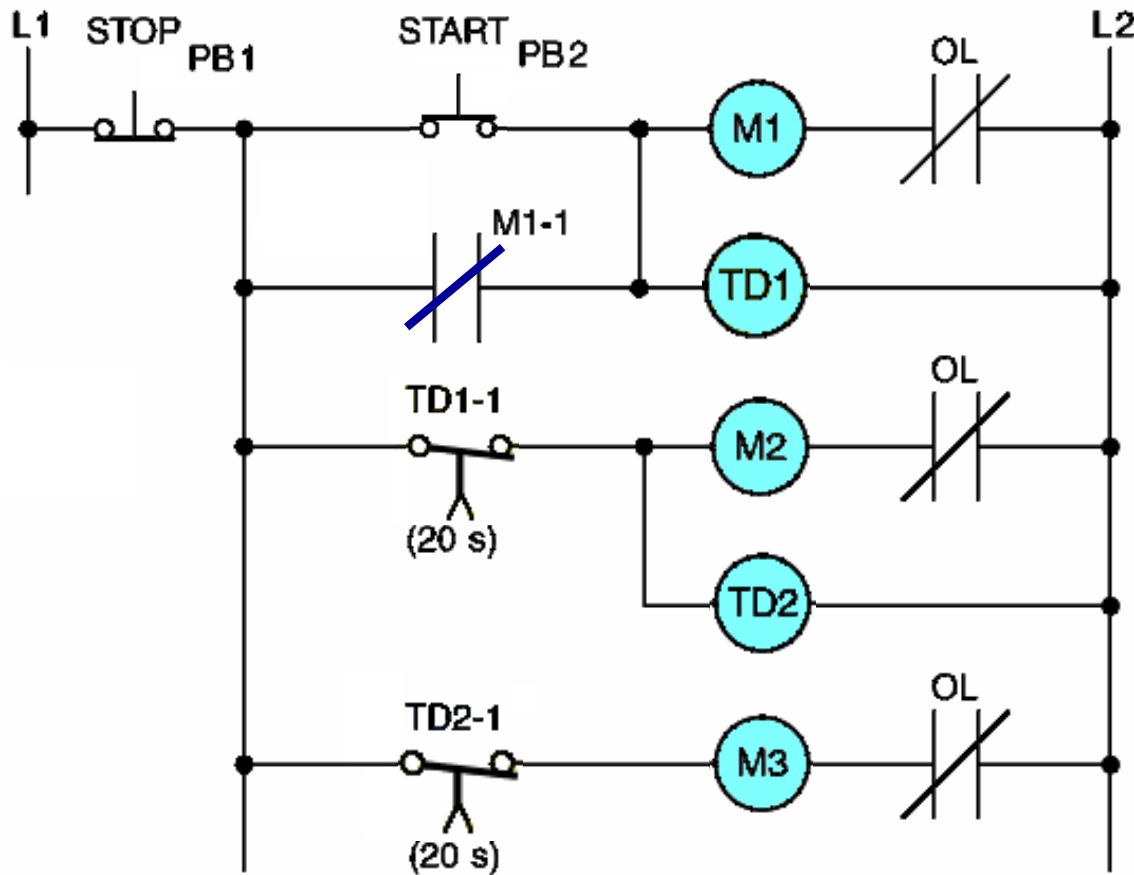
Bearing Lubrication Program

Ladder logic program



Cascading Timers

The programming of two or more timers together is called *cascading*. Timers may be interconnected, or cascaded to satisfy any required control logic.

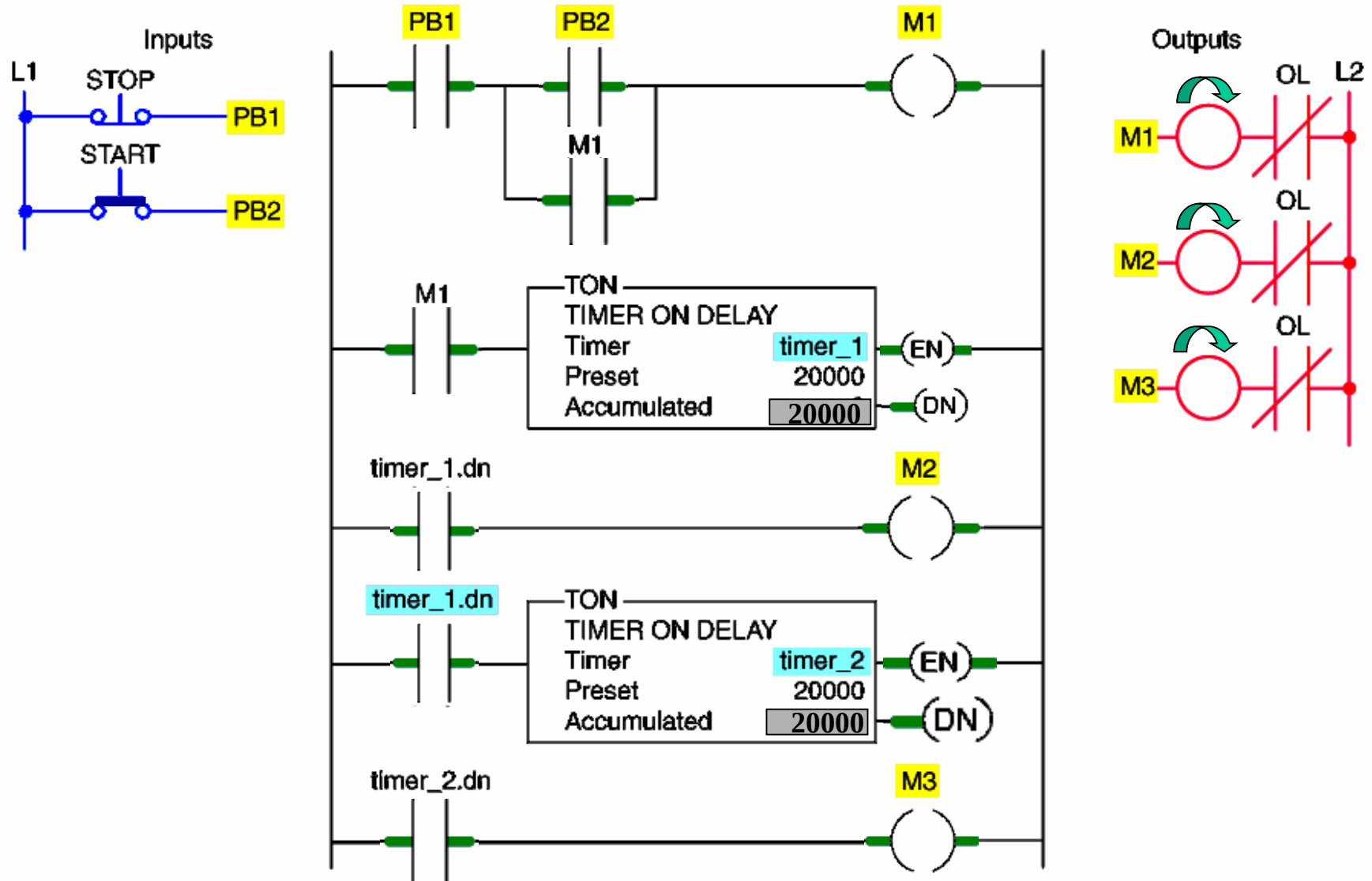


Relay Schematic Diagram

Three motors started automatically in sequence with a 20-s time delay between each motor startup.

Equivalent Time-Delayed Motor-Starting Program

Ladder logic program



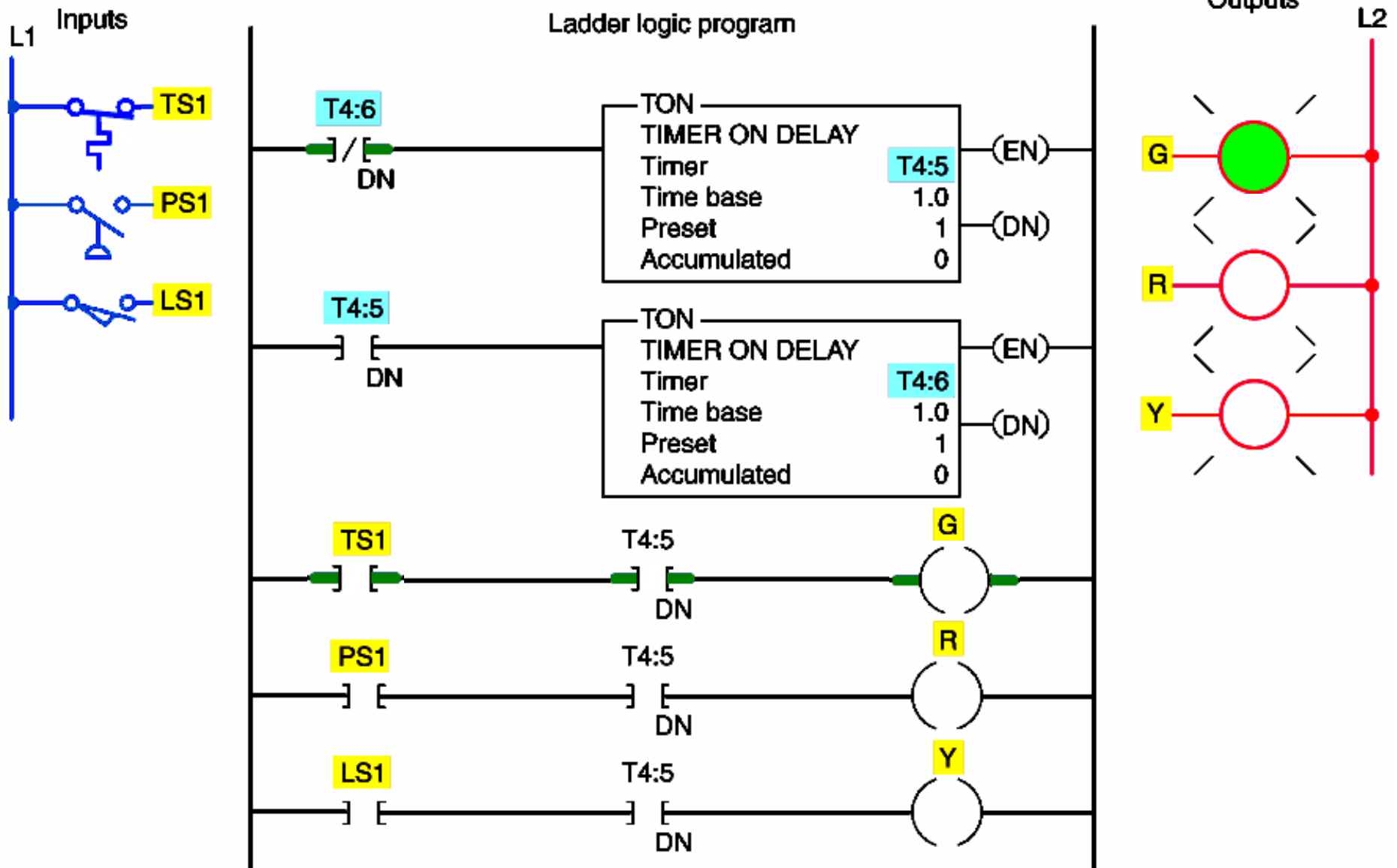
Annunciator Flasher Circuit

Two timers can be interconnected to form an oscillator circuit. The oscillator logic is basically a timing circuit programmed to generate periodic output pulses of any duration. They can be used as part of an annunciator system to indicate an alarm condition.

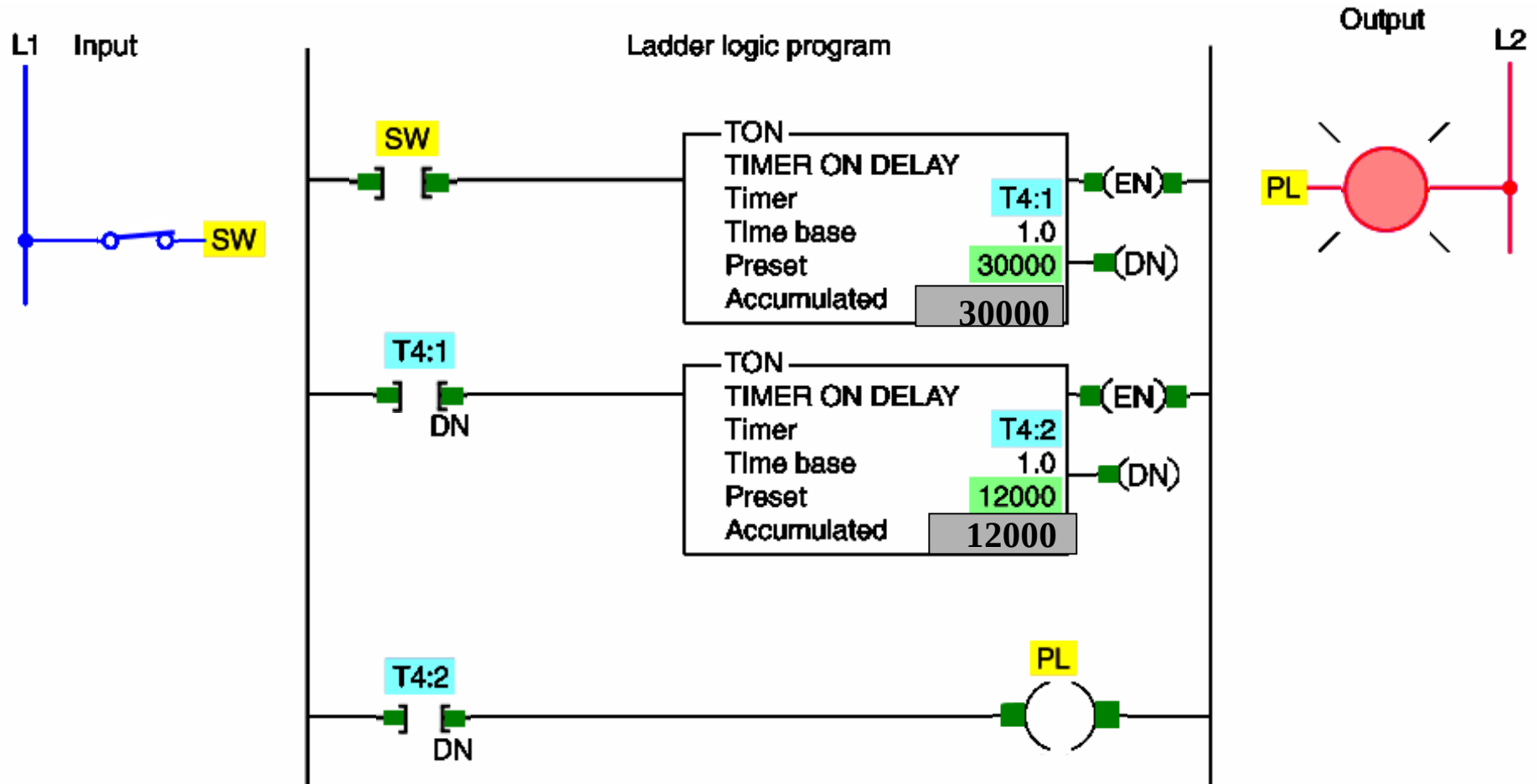
 white shutter open	 White shutter permit
 FE Vac valve open	 Beam line Vac Ok
 Mono shutter open	 Mono shutter permit

The oscillator circuit output is programmed in series with the alarm condition. If the alarm condition is true, the appropriate output indicating light will flash.

Annunciator Flasher Circuit

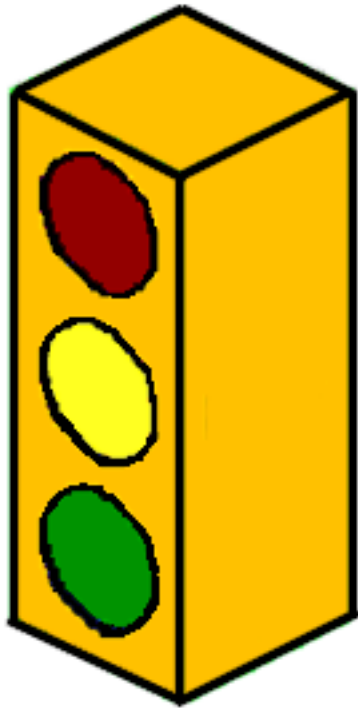


Cascading of Timers for Longer Time Delays

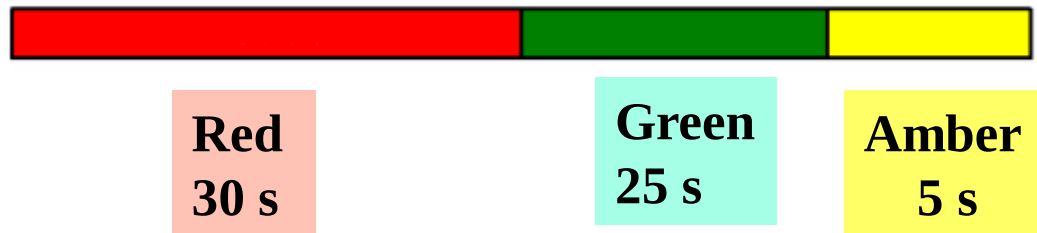


Control of Traffic Lights in One Direction

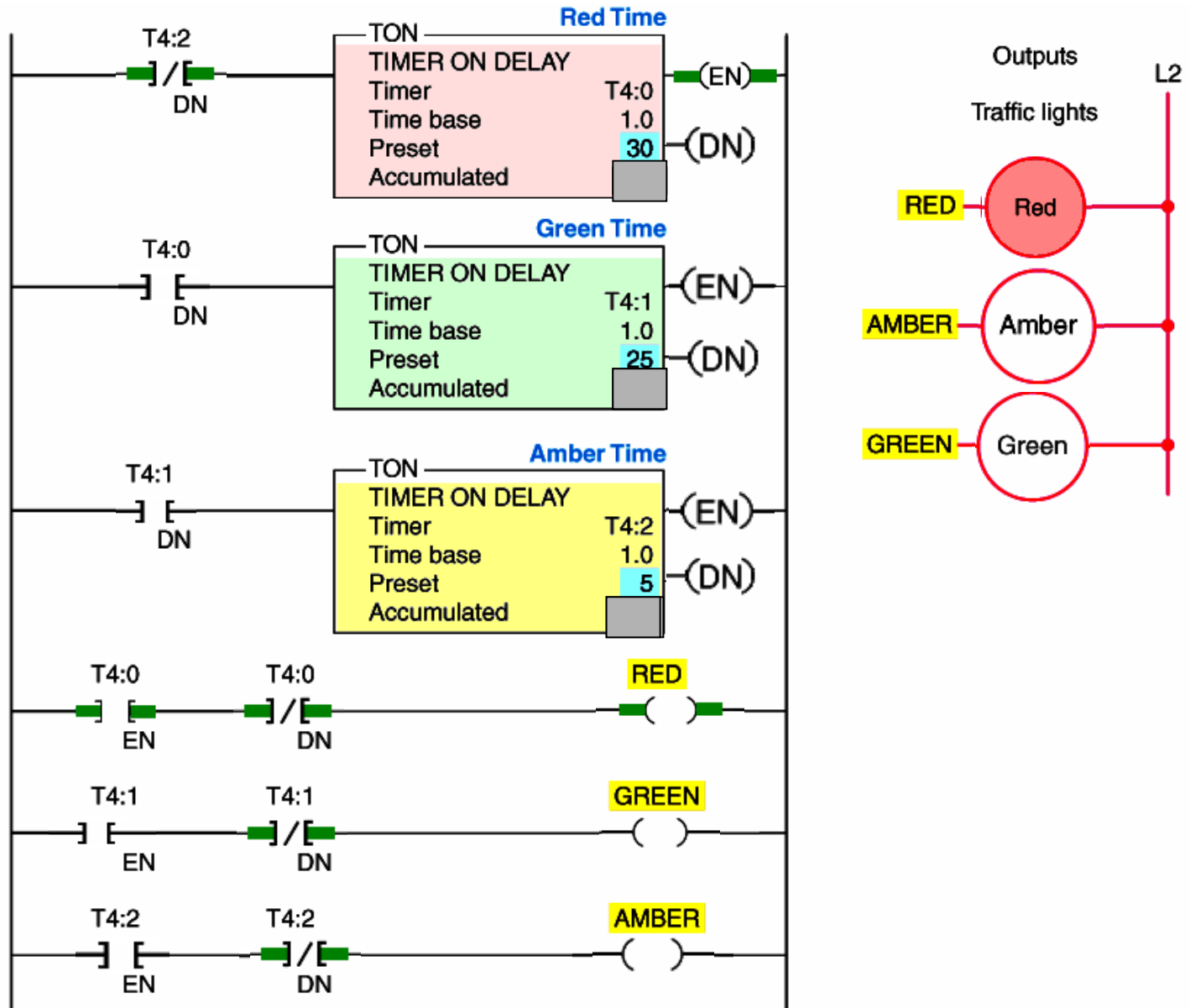
A typical application for PLC timers is the control of traffic lights.



Control of Traffic in One Direction Sequence of Operation



Control of Traffic Lights in One Direction



11. The timed contact of a PLC timer can only be used as a normally-open contact. (True/False)

12. The _____ bit operates the same as an instantaneous contact on a timer relay.

a. enable

b. done

c. timer-timing

d. timer number

13. The preset time of a PLC timer represents the amount of time that has elapsed from the moment the timing coil became energized. (True/False)

14. If the preset time of a timer is 150 and the time base is 0.1 seconds, the time-delay period would be 1500 seconds. (True/False)

15. In general, the three different types of PLC timers are:

a. TON, TOF, and PRE

b. TON, TOF, and RTO

c. TON, ACC, and RTO

d. TT, EN, and DN

16. The amount of time for which a timer is programmed is called the

a. preset

b. set point

c. Done Time (DN)

d. accumulated time

17. The timer reset instruction must be addressed to the same address as the _____ instruction.

a. TON

b. TOF

c. RTO

d. EN

18. Which of the following statements is *not* true for a retentive on-delay timer?

a. The timer accumulates time when it is energized.

b. The timer requires a reset instruction to reset the accumulated value of the timer to zero.

c. The timer does not reset to zero when it is de-energized.

d. The reset input to the timer will not override the control input.

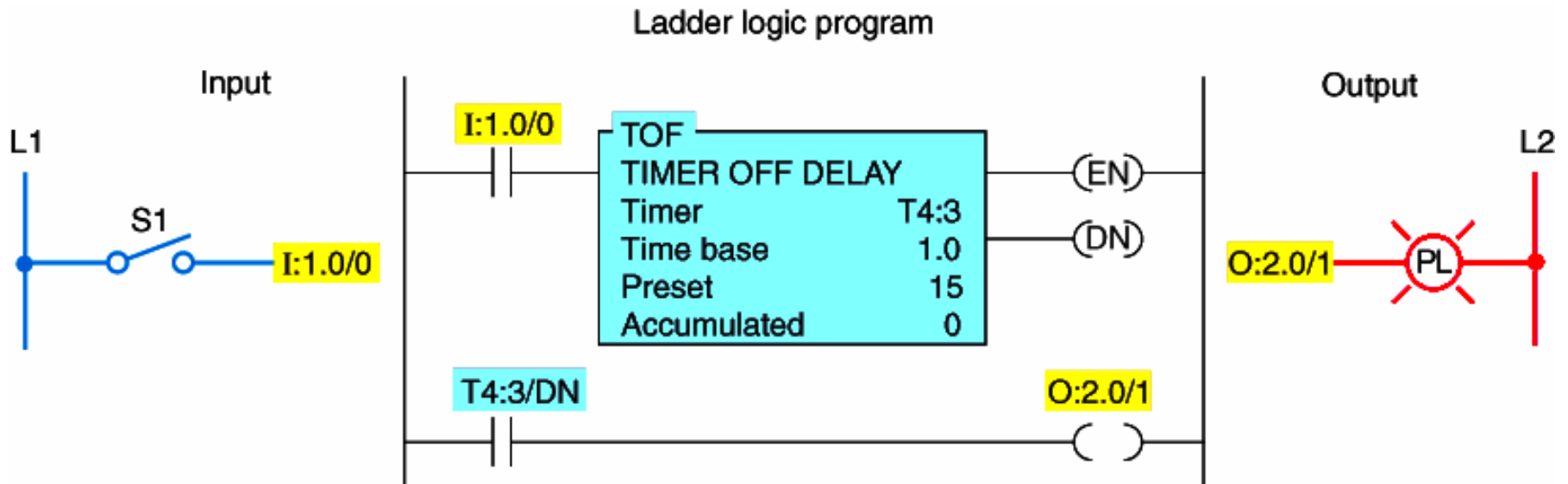
19. The timer instruction is:

- a. an input instruction
- b. an output instruction
- c. either a or b
- d. both a or b

20. The interconnection of timers is commonly called :

- a. grouping
- b. programming
- c. sequencing
- d. cascading

21. For the program shown, the pilot light will be:



- a. on at all times
- b. off at all times
- c. switched on 15 s after the switch has been actuated from the open to the closed position
- d. switched off 15 s after the switch has been actuated from the on to off position**